Ref. Ares(2023)2938080 - 26/04/2023

# Call: HORIZON-JU-SNS-2023

(HORIZON-JU-SNS-2023)

## Topic: HORIZON-JU-SNS-2023-STREAM-B-01-03

## Type of Action: HORIZON-JU-RIA

## Proposal number: 101139292

## Proposal acronym: Opti-6G

## Type of Model Grant Agreement: HORIZON Action Grant Budget-Based

## Table of contents


| Section | Title | Action |
|---|---|---|
| 1 | General information | |
| 2 | Participants | |
| 3 | Budget | |
| 4 | Ethics and security | |




This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

Proposal ID **101139292**

Acronym **Opti-6G**

# 1 - General information

Fields marked * are mandatory to fill.

| | | | |
|---|---|---|---|
| Topic | HORIZON-JU-SNS-2023-STREAM-B-01-03 | Type of Action | HORIZON-JU-RIA |
| Call | HORIZON-JU-SNS-2023 | Type of Model Grant Agreement | HORIZON-AG |

Acronym: Opti-6G

Proposal title: Optical 6G Cell Free Networks

Note that for technical reasons, the following characters are not accepted in the Proposal Title and will be removed: < > " &

Duration in months: 36

Fixed keyword 1: Communication engineering and systems telecommunications

Free keywords: *Cell free, optical, integration*

## Abstract *

The OPTI-6G project develops a Photonic near IR Cell Free 5G Network, which does not suffer from interference because of the propagation characteristics of EM waves in this part of the spectrum and provides universal broadband coverage within buildings from pervasively located OWC access points.
The benefits of applying cell free near IR networks in buildings are (1) multi-connectivity can be configured with cell free network thereby improving link quality and reliability; (2) there is no longer the need for building owners to subdivide their non-public mobile building network into cellular areas; (3) building owners no longer need to request permission from MNOs to use their licensed spectrum since the system operates at the optical unlicensed bands; (4) interference between inside and outside access is managed by an AI Based Distributed Scheduler; (5) position and orientation of end user equipment can be measured very accurately.
It provides solutions to two main barriers to develop this challenge: (i) Brings together a select multi-disciplinary team of research institutions and industries in a collaborative project to develop and demonstrate this vision, who otherwise would not have assembled to achieve this goal; (ii) Develops a proof of concept demonstrator in OLEDCOMM and RUNEL; (ii) Performs experiments to measure position and orientation of end user equipment at UVSQ and UBRU.
The starting point is (1) state-of-the-art cell free 5G RAN solution and very high accuracy sub cm accuracy localisation measurement system developed by RUNEL; (2) the state of the art VCSEL infrared source and high-speed PIN photodiodes receivers and that supports 1 Gbps data rates up to 5 m over a field of emission of 25° developed by OLEDCOMM.
The project brings together these technologies to produce an easy installation and operation, License Free broadband Network for indoor buildings. The outcome will be low-cost commercially attractive broadband cellular access within buildings

Remaining characters 0

Has this proposal (or a very similar one) been submitted in the past 2 years in response to a call for proposals under any EU programme, including the current call? ○ Yes ◉ No

| Please give the proposal reference or contract number. |
|---|
| *Previously submitted proposals should be with either 6 or 9 digits.* |



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# Administrative forms

Proposal ID **101139292**

Acronym **Opti-6G**

## Declarations

Field(s) marked * are mandatory to fill.

| | |
|---|---|
| 1) We declare to have the explicit consent of all applicants on their participation and on the content of this proposal. * | ☒ |
| 2) We confirm that the information contained in this proposal is correct and complete and that none of the project activities have started before the proposal was submitted (unless explicitly authorised in the call conditions). * | ☒ |
| 3) We declare:<br>- to be fully compliant with the eligibility criteria set out in the call<br>- not to be subject to any exclusion grounds under the EU Financial Regulation 2018/1046<br>- to have the financial and operational capacity to carry out the proposed project. * | ☒ |
| 4) We acknowledge that all communication will be made through the Funding & Tenders Portal electronic exchange system and that access and use of this system is subject to the Funding & Tenders Portal Terms and Conditions. * | ☒ |
| 5) We have read, understood and accepted the Funding & Tenders Portal Terms & Conditions and Privacy Statement that set out the conditions of use of the Portal and the scope, purposes, retention periods, etc. for the processing of personal data of all data subjects whose data we communicate for the purpose of the application, evaluation, award and subsequent management of our grant, prizes and contracts (including financial transactions and audits). * | ☒ |
| 6) We declare that the proposal complies with ethical principles (including the highest standards of research integrity as set out in the ALLEA European Code of Conduct for Research Integrity, as well as applicable international and national law, including the Charter of Fundamental Rights of the European Union and the European Convention on Human Rights and its Supplementary Protocols. Appropriate procedures, policies and structures are in place to foster responsible research practices, to prevent questionable research practices and research misconduct, and to handle allegations of breaches of the principles and standards in the Code of Conduct. * | ☒ |
| 7) We declare that the proposal has an exclusive focus on civil applications (activities intended to be used in military application or aiming to serve military purposes cannot be funded). If the project involves dual-use items in the sense of Regulation 2021/821, or other items for which authorisation is required, we confirm that we will comply with the applicable regulatory framework (e.g. obtain export/import licences before these items are used). * | ☒ |
| 8) We confirm that the activities proposed do not<br>- aim at human cloning for reproductive purposes;<br>- intend to modify the genetic heritage of human beings which could make such changes heritable (with the exception of research relating to cancer treatment of the gonads, which may be financed), or<br>- intend to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer.<br>- lead to the destruction of human embryos (for example, for obtaining stem cells)<br>These activities are excluded from funding. * | ☒ |
| 9) We confirm that for activities carried out outside the Union, the same activities would have been allowed in at least one EU Member State. * | ☒ |

The coordinator is only responsible for the information relating to their own organisation. Each applicant remains responsible for the information declared for their organisation. If the proposal is retained for EU funding, they will all be required to sign a declaration of honour.

**False statements** or incorrect information may lead to administrative sanctions under the EU Financial Regulation.



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Proposal ID **101139292**

Acronym **Opti-6G**

# 2 - Participants

## List of participating organisations

| # | Participating Organisation Legal Name | Country | Role | Action |
|---|---|---|---|---|
| 1 | EURESCOM-EUROPEAN INSTITUTE FOR RESEARCH AND ST | DE | Coordinator | |
| 2 | BRUNEL UNIVERSITY LONDON | United Kingdom | Partner | |
| 3 | UNIVERSITE DE VERSAILLES SAINT-QUENTIN-EN-YVELINES | FR | Partner | |
| 4 | RUNEL NGMT LTD | IL | Partner | |
| 5 | OLEDCOMM SAS | FR | Partner | |



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# Organisation data

| PIC | Legal name |
|---|---|
| *999912279* | *EURESCOM-EUROPEAN INSTITUTE FOR RESEARCH AND STRATEGIC STUDIES IN TELECOMMUNICATIONS GMBH* |

Short name: EUR

Address

| | |
|---|---|
| Street | WIEBLINGER WEG 19/4 |
| Town | HEIDELBERG |
| Postcode | 69123 |
| Country | Germany |
| Webpage | www.eurescom.eu |

## Specific Legal Statuses

| | |
|---|---|
| Legal person | yes |
| Public body | no |
| Non-profit | no |
| International organisation | no |
| Secondary or Higher education establishment | no |
| Research organisation | no |

**SME Data**

**Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.**

| | | |
|---|---|---|
| SME self-declared status | 25/09/2020 | - no |
| SME self-assessment | 21/08/2008 | - no |
| SME validation | 21/08/2008 | - no |



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# Administrative forms

## Departments carrying out the proposed work

**Department 1**

| | | |
|---|---|---|
| Department name | Programme Management | ☐ not applicable |
| | ☒ Same as proposing organisation's address | |
| Street | WIEBLINGER WEG 19/4 | |
| Town | HEIDELBERG | |
| Postcode | 69123 | |
| Country | Germany | |

## Links with other participants

| Type of link | Participant |
|---|---|



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## Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Ádám**

Last name* **KAPOVITS**

E-Mail* **kapovits@eurescom.eu**

Position in org. Programme Manager

Department Programme Management

☐ Same as organisation name

☒ Same as proposing organisation's address

Street WIEBLINGER WEG 19/4

Town HEIDELBERG

Post code 69123

Country Germany

Website *Please enter website*

Phone *+xxx xxxxxxxxx*

*Phone 2* *+xxx xxxxxxxxx*

## Other contact persons

| First Name | Last Name | E-mail | Phone |
|---|---|---|---|
| David | Kennedy | kennedy@eurescom.eu | +xxx xxxxxxxxx |
| Luitgard | Hauer | hauer@eurescom.eu | +xxx xxxxxxxxx |
| Ellen | Tallas | tallas@eurescom.eu | +xxx xxxxxxxxx |



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## Researchers involved in the proposal

| Title | First Name | Last Name | Gender | Nationality | E-mail | Career Stage | Role of researcher (in the project) | Reference Identifier | Type of identifier |
|---|---|---|---|---|---|---|---|---|---|
| Mr | Adam | Kapovits | Man | Hungary | kapovits@eurescom.eu | Category B  Senior resea | Leading | 0000-0002-1833-8937 | Orcid ID |



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## Role of participating organisation in the project

| | |
|---|---|
| Project management | ☒ |
| Communication, dissemination and engagement | ☒ |
| Provision of research and technology infrastructure | ☐ |
| Co-definition of research and market needs | ☒ |
| Civil society representative | ☐ |
| Policy maker or regulator, incl. standardisation body | ☐ |
| Research performer | ☐ |
| Technology developer | ☐ |
| Testing/validation of approaches and ideas | ☐ |
| Prototyping and demonstration | ☐ |
| IPR management incl. technology transfer | ☒ |
| Public procurer of results | ☐ |
| Private buyer of results | ☐ |
| Finance provider (public or private) | ☐ |
| Education and training | ☐ |
| Contributions from the social sciences or/and the humanities | ☐ |
| Other<br>If yes, please specify: (Maximum number of characters allowed: 50) | ☒ |

Liaison with SNS JU



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**List of up to 5 publications,** widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

| Type of achievement | Short description (Max 500 characters) |
|---|---|
| *Publication* | *Bahare, Massod Khorsandi, Gavras, Anastasius, Gramaglia, Marco, Cosmas, John, Li, Xi, Bulakci, Ömer, Rahman, Arifur, Kostopoulos, Alexandros, Mesodiakaki, Agapi, Tsolkas, Dimitris, Ericson, Mårten, Boldi, Mauro, Uusitalo, Mikko, Ghoraishi, Mir, & Rugeland, Patrik. (2023). The 6G Architecture Landscape - European perspective. Zenodo. https://doi.org/10.5281/zenodo.7313232* |
| *Publication* | *Nielsen, Lars, Gavras, Anastasius, Dieudonne, Michael, Mesogiti, Ioanna, Roosipuu, Priit, Houatra, Drissa, & Kosmatos, Evangelos. (2022). Beyond 5G/6G KPIs and Target Values. Zenodo. https://doi.org/10.5281/zenodo.6577506* |
| *Publication* | *Wikström, Gustav, Schuler Scott, Arianna, Mesogiti, Ioanna, Stoica, Razvan-Andrei, Georgiev, Georgi, Barmpounakis, Sokratis, Gavras, Anastasius, Demestichas, Panagiotis, Hamon, Marie-Helene, Hallingby, Hanne-Stine, & Lund, David. (2022). What societal values will 6G address?. Zenodo. https://doi.org/10.5281/zenodo.6557534* |
| *Publication* | *Hallingby, Hanne Kristine, Fletcher, Simon, Frascolla, Valerio, Gavras, Anastasius, Mesogiti, Ioanna, & Parzysz, Fanny. (2021). 5G Ecosystems. Zenodo. https://doi.org/10.5281/zenodo.5094340* |
| *Publication* | *Kaloxylos, Alexandros, Gavras, Anastasius, Camps Mur, Daniel, Ghoraishi, Mir, & Hrasnica, Halid. (2020). AI and ML – Enablers for Beyond 5G Networks. Zenodo. https://doi.org/10.5281/zenodo.4299895* |

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

| Name of Project or Activity | Short description (Max 500 characters) |
|---|---|
| *6G-BRAINS (H2020 - 101017226)* | *6G BRAINS addresses current challenges in networks, beyond 5G (B5G), 6G, Terahertz (THz),Optical Wireless Communications (OWC), Artificial Intelligence (AI), Multi-agent Deep Reinforcement Learning, Highly Dynamic Ultra-Dense Device-to-Device (D2D) Cell-free Networks, Grant Free-Non-Orthogonal Multiple Access (GF-NOMA), End-to-End (E2E) Slicing, Integrated Access Backhaul (IAB), Industrial Virtual Assistant (IVA), Simultaneous Localization and Mapping (SLAM), Content Distribution Network (CDN).* |
| *IoRL - Internet of Radio Light (H2020 - 761992)* | *The Internet of Radio Light project (July 2017-Dec 2020) was a 5G PPP phase 2 project that researched and developed a 5G solution integrating visible light and mmWave for improved 5G indoor connectivity and location accuracy.* |

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

| Name of infrastructure of equipment | Short description (Max 300 characters) |
|---|---|
| *EuresTools* | *Eurescom offers Cloud-based tools for effective project management under the brand name EuresTools®. More than 250 European research projects have used EuresTools® to work effectively and comply with EC reporting and auditing requirements. The tool suite is hosted in Germany and is GDPR compliant* |



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## Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

**Minimum process-related requirements (building blocks) for a GEP**

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.



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| PIC | Legal name |
|---|---|
| *999749610* | *BRUNEL UNIVERSITY LONDON* |

## Short name: UBRU

Address

| | |
|---|---|
| Street | KINGSTON LANE |
| Town | UXBRIDGE |
| Postcode | UB8 3PH |
| Country | United Kingdom |
| Webpage | www.brunel.ac.uk |

## Specific Legal Statuses

| | |
|---|---|
| Legal person | yes |
| Public body | yes |
| Non-profit | yes |
| International organisation | no |
| Secondary or Higher education establishment | yes |
| Research organisation | yes |

**SME Data**

**Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.**

| | |
|---|---|
| SME self-declared status | 26/02/2014 - no |
| SME self-assessment | unknown |
| SME validation | unknown |



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## Departments carrying out the proposed work

**Department 1**

| | | |
|---|---|---|
| Department name | Electronic and Electrical Engineering, College of Engineering and | ☐ not applicable |
| | ☒ Same as proposing organisation's address | |
| Street | KINGSTON LANE | |
| Town | UXBRIDGE | |
| Postcode | UB8 3PH | |
| Country | United Kingdom | |

## Links with other participants

| Type of link | Participant |
|---|---|



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## Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Prof.

Gender ◯ Woman ◉ Man ◯ Non Binary

First name* **John**

Last name* **Cosmas**

E-Mail* **john.cosmas@brunel.ac.uk**

Position in org. Research Professor

Department College of Engineering, Design and Physical Sciences

☐ Same as organisation name

☒ Same as proposing organisation's address

Street KINGSTON LANE

Town UXBRIDGE

Post code UB8 3PH

Country United Kingdom

Website https://www.brunel.ac.uk/

Phone +44 1895266756

Phone 2 +447908151859



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## Researchers involved in the proposal

| Title | First Name | Last Name | Gender | Nationality | E-mail | Career Stage | Role of researcher (in the project) | Reference Identifier | Type of identifier |
|---|---|---|---|---|---|---|---|---|---|
| Prof | John | Cosmas | Man | United Kingdom | john.cosmas@brunel.ac.uk | Category A  Top grade re | Leading | 0000-0003-4378-5576 | Orcid ID |
| Prof | Hamed | Al-Raweshidy | Man | United Kingdom | Hamed.al-raweshidy@brunel.ac.uk | Category A  Top grade re | Team member | 0000-0002-3702-8192 | Orcid ID |
| Dr | Nila | Nilavalan | Man | United Kingdom | nila.nilavalan@brunel.ac.uk | Category A  Top grade re | Team member | 0000-0001-8168-2039 | Orcid ID |



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## Role of participating organisation in the project

| | |
|---|---|
| Project management | ☒ |
| Communication, dissemination and engagement | ☒ |
| Provision of research and technology infrastructure | ☒ |
| Co-definition of research and market needs | ☒ |
| Civil society representative | ☐ |
| Policy maker or regulator, incl. standardisation body | ☐ |
| Research performer | ☒ |
| Technology developer | ☒ |
| Testing/validation of approaches and ideas | ☒ |
| Prototyping and demonstration | ☒ |
| IPR management incl. technology transfer | ☒ |
| Public procurer of results | ☐ |
| Private buyer of results | ☐ |
| Finance provider (public or private) | ☐ |
| Education and training | ☒ |
| Contributions from the social sciences or/and the humanities | ☐ |
| Other<br>If yes, please specify: (Maximum number of characters allowed: 50) | ☐ |



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**List of up to 5 publications,** widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

| Type of achievement | Short description (Max 500 characters) |
|---|---|
| *Publication* | *Agapi Mesodiakaki, Alexandros Kostopoulos, Anastasius Gavras, Arifur Rahman, Bahare Masood Khorsandi, Dimitris Tsolkas, John Cosmas, Marco Gramaglia, Mårten Ericson, Mauro Boldi, Mikko Uusitalo, Mir Ghoraishi, Ömer Bulakci, Patrik Rugeland, Xi Li, et al. "Towards Sustainable and Trustworthy 6G System Challenges, Enablers, and Architectural Design" Wiley Book, June 2023.* |
| *Publication* | *Ben Meunier, John Cosmas, Kareem Ali, Nawar Jawad, Geoffrey Eappen, Wei Li, Hequn Zhang "Visible Light Positioning with Lens Compensation for Non-Lambertian Emission" IEEE Transactions on Broadcasting, March 2023* |
| *Publication* | *Geoffrey Eappen, John Cosmas, Shankar T, Rajesh A, Nilavalan R, and Joji Thomas "Deep Learning Integrated Reinforcement Learning for Adaptive Beamforming in B5G Networks" Provisionally IET Communications Journals, November 2022* |
| *Publication* | *Kareem Ali , John Cosmas, Yue Zhang , Ben Meunier, Nawar Jawad, Xun Zhang, Lina Shi "Measurement Campaign on 5G Indoor millimeter Wave and Visible Light Communications Multi Component Carrier System" IEEE Transactions on Broadcasting (Volume: 68, Issue: 1, March 2022* |
| *Other achievement* | *Horizon Project Innovation Radar Award for the ultra accurate User Equipment (UE) location measurement algorithm ( 1cm) developed in the 6G Brains project* |

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

| Name of Project or Activity | Short description (Max 500 characters) |
|---|---|
| *EU Horizon 2020, 6G BRAINS* | *"Bringing Reinforcement-learning Resource Allocation of Radio Light Network for Massive Connections" Jan 2021 – Dec 2023: The design of 6G networks for Industry 4.0 with the support of Artificial Intelligence and Reinforcement Learning.* |
| *EU, Horizon 2020, IoRL* | *"Internet of Radio Light" July 2017-Dec 2020: developed a safer, more secure, customisable and intelligent 5G building network using millimetre wave (mmWave) and Visible Light Communications (VLC).* |
| *EU FP6, IST PLUTO* | *"Physical Layer DVB Transmission Optimisation" Jan 2006 – June 2008: Reliable transmission of Digital TV and Internet networks services using MIMO.* |
| *EU FP6, IST INSTINCT* | *"IP-based Networks, Services and TermINals for Converging SysTems" Jan 2004 – Dec 2005: The design of Convergent Broadcast TV and Internet networks* |
| *EU, FP5, IST CISMUMDUS* | *"Convergence of IP services for mobile users in DVB and UMTS networks" July 2001 – Dec, 2003: The design of convergent mobile terminals with broadcast TV receiver.* |

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

| Name of infrastructure of equipment | Short description (Max 300 characters) |
|---|---|
| *5G Lab* | *5G RAN sub 6GHz on-loan from RUNEL, to measure distance and location to 1cm accuracy.<br>Also includes<br>IR OWC from OLEDCOMM,<br>2 Alienware Gaming Laptops,<br>2 Linux Raspberry Pi / Snapdragon<br>Ouster Lidar,<br>Ouznest flatbed x, y, z plotter,<br>Theta V 360 degree camera,<br>etc* |



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| | |
|---|---|
| *Wireless Networks and Communications Lab* | *General purpose PhD student lab which includes*<br>*Private Openstack Linux Multiaccess edge Computing Cloud*<br>*USRP Workstation*<br>*Rexel Industrial Programable Logic Controller,*<br>*1 GHz Spectrum Analyser*<br>*etc* |
| *Wireless Computer Communications Lab* | *5G/WLAN Linux Raspberry Pi IoT lab with 100 workstations used to develop applications such as remote / autonomous driving of vehicles and drones beyond the line of sight of operators and recognition of objects using Artificial Intelligence from captured video.,*<br>*wide range of peripheral devices, etc* |



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## Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

**Minimum process-related requirements (building blocks) for a GEP**

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.



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| PIC | Legal name |
|---|---|
| *999837104* | *UNIVERSITE DE VERSAILLES SAINT-QUENTIN-EN-YVELINES.* |

## Short name: UVSQ

Address

| | |
|---|---|
| Street | Avenue de Paris 55 |
| Town | VERSAILLES |
| Postcode | 78035 |
| Country | France |
| Webpage | www.uvsq.fr |

## Specific Legal Statuses

| | |
|---|---|
| Legal person | yes |
| Public body | yes |
| Non-profit | yes |
| International organisation | no |
| Secondary or Higher education establishment | yes |
| Research organisation | yes |

**SME Data**

**Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.**

| | |
|---|---|
| SME self-declared status | 21/01/2022 - no |
| SME self-assessment | unknown |
| SME validation | unknown |



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## Departments carrying out the proposed work

**Department 1**

| | | |
|---|---|---|
| Department name | LISV (Laboratoire d'Ingénierie des Systèmes de Versailles) | ☐ not applicable |
| | ☐ Same as proposing organisation's address | |
| Street | 10-12 avenue de l'Europe | |
| Town | Vélizy-Villacoublay | |
| Postcode | 78140 | |
| Country | France | |

## Links with other participants

| Type of link | Participant |
|---|---|



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## Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Dr

Gender: ○ Woman ◉ Man ○ Non Binary

First name*: **Bastien**

Last name*: **Béchadergue**

E-Mail*: **bastien.bechadergue@uvsq.fr**

Position in org.: Associate professor

Department: LISV

☐ Same as organisation name

☐ Same as proposing organisation's address

Street: 10-12 avenue de l'Europe

Town: Vélizy-Villacoublay

Post code: 78140

Country: France

Website: https://www.lisv.uvsq.fr/

Phone: +33139254951

*Phone 2*: *+xxx xxxxxxxxx*

## Other contact persons

| First Name | Last Name | E-mail | Phone |
|---|---|---|---|
| Luc | Chassagne | luc.chassagne@uvsq.fr | +33607714617 |
| Estelle | Tronci | estelle.tronci@uvsq.fr | +xxx xxxxxxxxx |



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

## Researchers involved in the proposal

| Title | First Name | Last Name | Gender | Nationality | E-mail | Career Stage | Role of researcher (in the project) | Reference Identifier | Type of identifier |
|---|---|---|---|---|---|---|---|---|---|
| Dr | Bastien | BECHADERGUE | Man | France | bastien.bechadergue@uvsq.fr | Category B  Senior resea | Leading | 0000-0002-0867-4865 | Orcid ID |
| Prof | Luc | CHASSAGNE | Man | France | luc.chassagne@uvsq.fr | Category A  Top grade re | Team member | 0000-0003-4240-4591 | Orcid ID |
| Dr | Barthélemy | CAGNEAU | Man | France | barthelemy.cagneau@uvsq.fr | Category B  Senior resea | Team member | 0000-0002-6285-7929 | Orcid ID |
| Dr | Hongyu | GUAN | Man | China (People's | hongyu.guan@uvsq.fr | Category B  Senior resea | Team member | 0000-0003-1220-6834 | Orcid ID |



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

# Administrative forms

## Role of participating organisation in the project

| | |
|---|---|
| Project management | ☐ |
| Communication, dissemination and engagement | ☒ |
| Provision of research and technology infrastructure | ☒ |
| Co-definition of research and market needs | ☒ |
| Civil society representative | ☐ |
| Policy maker or regulator, incl. standardisation body | ☐ |
| Research performer | ☒ |
| Technology developer | ☒ |
| Testing/validation of approaches and ideas | ☒ |
| Prototyping and demonstration | ☒ |
| IPR management incl. technology transfer | ☐ |
| Public procurer of results | ☐ |
| Private buyer of results | ☐ |
| Finance provider (public or private) | ☐ |
| Education and training | ☒ |
| Contributions from the social sciences or/and the humanities | ☐ |
| Other<br>If yes, please specify: (Maximum number of characters allowed: 50) | ☐ |



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

**List of up to 5 publications,** widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

| Type of achievement | Short description (Max 500 characters) |
|---|---|
| *Publication* | *H. Al Satai, B. Béchadergue, L. Chassagne, W. M. Ridha Shakir, "Coverage Optimization With Beamsteering-Based Indoor Optical Wireless Communications", IEEE International Conference on Communications (ICC): Optical Networks and Systems Symposium 2023 (Rome, Italy), June 2023.* |
| *Publication* | *L. Shi, B. Béchadergue, L. Chassagne, and H. Guan, "Joint Visible Light Sensing and Communication Using m-CAP Modulation", IEEE Transactions on Broadcasting, vol. 69, no. 1, pp. 276-288, March 2023. DOI: 10.1109/TBC.2022.3201649.* |
| *Publication* | *H. Zhao, J. Wang, X. Zhang, and B. Béchadergue, "A High-Precision Positioning Scheme under Non-Point Visible Transmitters", IEEE Open Journal of the Communications Society, vol. 1, pp. 1131-1139, July 2020. DOI: 10.1109/OJCOMS.2020.3008340* |
| *Publication* | *B. Béchadergue, L. Chassagne, and H. Guan, "Simultaneous visible light communication and distance measurement based on the automotive lighting," IEEE Transactions on Intelligent Vehicles, vol. 4, no. 4, pp. 532-547, Dec. 2019. DOI: 10.1109/TIV.2019.2938087* |
| *Publication* | *M. Mohammedi Merah, H. Guan, L. Chassagne, "Experimental Multi-User Visible Light Communication Attocell using Multiband Carrierless Amplitude and Phase Modulation", IEEE Access, vol. 7, pp 12742-12754, Jan. 2019. DOI: 10.1109/ACCESS.2019.2893451* |

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

| Name of Project or Activity | Short description (Max 500 characters) |
|---|---|
| *SAFELiFi (French ANR agency)* | *2021-2024: Steerable Light Antenna for Enhanced QoS Cell-Free LiFi Networks: Development of beamsteering solutions for cell free LiFi networks. Simulation and modelling of the parameters, design of steerable antenna and new method for sensing of the receiver position and orientation.* |
| *SOLERIS (French AID agency)* | *2022-2024: Solutions LiFi sécurisées durcies: Development of beamsteering solutions with sensing implementation in the context of a for cell-free network deployment for headquarters.* |

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

| Name of infrastructure of equipment | Short description (Max 300 characters) |
|---|---|
| *LiFi Lab (LISV)* | *4 digital generators (high frequency signals)*<br>*1 network analyser (up to 8 GHz)*<br>*3 spectrum analyzers (up to 7 GHz)*<br>*Optical antennas (from OLEDCOMM)*<br>*Specific room dedicated to OWC/LiFi experiments*<br>*2 high frequency digital oscilloscope*<br>*Matlab software licence for simulation*<br>*3 PC for test bench control* |



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

## Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

**Minimum process-related requirements (building blocks) for a GEP**

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

| PIC | Legal name |
|---|---|
| *934564543* | *RUNEL NGMT LTD* |

## Short name: REL

Address

| | |
|---|---|
| Street | 2 HACHOMA STREET ECOND FLOOR |
| Town | RISHON LETZION |
| Postcode | 7565502 |
| Country | Israel |
| Webpage | www.runel.net |

## Specific Legal Statuses

| | |
|---|---|
| Legal person | yes |
| Public body | no |
| Non-profit | no |
| International organisation | no |
| Secondary or Higher education establishment | no |
| Research organisation | no |

**SME Data**

**Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.**

| | | |
|---|---|---|
| SME self-declared status | 31/12/2018 | - yes |
| SME self-assessment | 31/12/2018 | - yes |
| SME validation | unknown | |



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

## Departments carrying out the proposed work

**No department involved**

Department name *Name of the department/institute carrying out the work.* ☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

## Links with other participants

| Type of link | Participant |
|---|---|



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## Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ◯ Woman ◉ Man ◯ Non Binary

First name* **Israel**

Last name* **Koffman**

E-Mail* **israelk@runel.net**

Position in org. Executive Vice President

Department RUNEL NGMT LTD

☒ Same as organisation name

☒ Same as proposing organisation's address

Street 2 HACHOMA STREET ECOND FLOOR

Town RISHON LETZION

Post code 7565502

Country Israel

Website www.runel.net

Phone +972545303110

*Phone 2* *+xxx xxxxxxxxx*

## Other contact persons

| First Name | Last Name | E-mail | Phone |
|---|---|---|---|
| Zion | Hadad | zionh@runel.net | +972544560655 |



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

## Researchers involved in the proposal

| Title | First Name | Last Name | Gender | Nationality | E-mail | Career Stage | Role of researcher (in the project) | Reference Identifier | Type of identifier |
|---|---|---|---|---|---|---|---|---|---|
| Dr | Zion | Hadad | Man | Israel | zionh@runel.net | Category A  Top grade re | Leading | 0009-0004-3270-7630 | Orcid ID |
| Mr | Baruch | Globen | Man | Israel | baruchg@runel.net | Category B  Senior resea | Team member | 0009-0000-6942-3157 | Orcid ID |
| Mrs | Israel | Koffman | Man | Israel | israelk@runel.net | Category B  Senior resea | Team member | 0009-0004-9558-0278 | Orcid ID |



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

# Administrative forms

## Role of participating organisation in the project

| | |
|---|---|
| Project management | ☐ |
| Communication, dissemination and engagement | ☒ |
| Provision of research and technology infrastructure | ☒ |
| Co-definition of research and market needs | ☒ |
| Civil society representative | ☐ |
| Policy maker or regulator, incl. standardisation body | ☐ |
| Research performer | ☒ |
| Technology developer | ☒ |
| Testing/validation of approaches and ideas | ☒ |
| Prototyping and demonstration | ☒ |
| IPR management incl. technology transfer | ☐ |
| Public procurer of results | ☐ |
| Private buyer of results | ☐ |
| Finance provider (public or private) | ☐ |
| Education and training | ☐ |
| Contributions from the social sciences or/and the humanities | ☐ |
| Other<br>If yes, please specify: (Maximum number of characters allowed: 50) | ☐ |



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**List of up to 5 publications,** widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

| Type of achievement | Short description (Max 500 characters) |
|---|---|
| *Good* | *Development of the Sparq-2025P 5G end to end system for private network specially designed for applications that needs low latency and accurate UE position measurements such as : Industry 4.0; Smart Transportation, AR/VR/XR, Robotics, Drones, Remote Surgery, etc* |
| *Publication* | *RunEL participation ( Israel Koffman) in cowriting the 5GPP Technology Board document :"Non-Public-Networks – State of the art and way forward" - URL: http://doi.org/10.5281/zenodo.7230191* |
| *Publication* | *RunEL CEO (Dr Zion Hadad) Presentation in the 2022 Workshop on Machine Learning for Communications on Advanced Cellular Networks* |
| *Other achievement* | *Horizon Project Innovation Radar Award for the ultra accurate User Equipment (UE) location measurement algorithm ( 1cm) developed in the 6G Brains project* |
| *Other achievement* | *Horizon Project Innovation Radar Award for the ultra accurate User Equipment (UE) location measurement algorithm ( 1cm) developed in the 6G Brains project* |

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

| Name of Project or Activity | Short description (Max 500 characters) |
|---|---|
| *IORL- Horizon 2020* | *Development of 5G Infrastructure system in sub 6GHz band and in the Visual Light Communication ( VLC) Band* |
| *5 Genesis - Horizon 2020* | *Enhancing and testing the 5G Infrustructure system to achieve low end to end latency in the 1 millisecond range.* |
| *Affordable 5G - Horizon 2020* | *Enhancing the 5G Infrastructure system to include Industry standard Open Radio Access Interface (ORAN) to the Radio Unit ( RU)* |
| *6G Brain - Horizon 2020* | *Enhancing the 5G Infrastructure system to include an ultra accurate User Equipment ( UE) location measurement algorithm in the one centimeter range* |
| *Soccer Stadium Project* | *Installing a commercial 5G Network in a soccer stadium in Israel that includes the low end to end latency ( 1 msec) and high position UE location accuracy ( 1 cm) features* |

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

| Name of infrastructure of equipment | Short description (Max 300 characters) |
|---|---|
| *Sparq-2025P* | *The Sparq-2025P is a complete 5G infrastructure system for private network that is composed of 2 main modules: a) the NIB ( Network in a Box) including the 5G Core, CU , DU and application Server on site and b) the Radio Units (RU) at sub 6GHz band* |



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## Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

**Minimum process-related requirements (building blocks) for a GEP**

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

| PIC | Legal name |
|---|---|
| *916612074* | *OLEDCOMM SAS* |

## Short name: OLED

Address

| | |
|---|---|
| Street | 10-12 AVENUE DE L'EUROPE |
| Town | VELIZY-VILLACOUBLAY |
| Postcode | 78140 |
| Country | France |
| Webpage | www.oledcomm.net |

## Specific Legal Statuses

| | |
|---|---|
| Legal person | yes |
| Public body | no |
| Non-profit | no |
| International organisation | no |
| Secondary or Higher education establishment | no |
| Research organisation | no |

**SME Data**

**Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.**

| | | |
|---|---|---|
| SME self-declared status | 31/12/2019 | - yes |
| SME self-assessment | 31/12/2019 | - yes |
| SME validation | 31/12/2020 | - yes |



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

## Departments carrying out the proposed work

**No department involved**

Department name *Name of the department/institute carrying out the work.* ☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

## Links with other participants

| Type of link | Participant |
|---|---|



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## Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ◯ Woman ◉ Man ◯ Non Binary

First name* **Benjamin**

Last name* **Azoulay**

E-Mail* **benjamin.azoulay@oledcomm.com**

Position in org. President and CEO

Department OLEDCOMM SAS

☒ Same as organisation name

☒ Same as proposing organisation's address

Street 10-12 AVENUE DE L'EUROPE

Town VELIZY-VILLACOUBLAY

Post code 78140

Country France

Website www.oledcomm.net

Phone +33 7 85 14 72 64

*Phone 2* *+xxx xxxxxxxxx*

## Other contact persons

| First Name | Last Name | E-mail | Phone |
|---|---|---|---|
| Banjamin | Azoulay | benjamin.azoulay@oledcomm.net | +33 7 85 14 72 64 |



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

## Researchers involved in the proposal

| Title | First Name | Last Name | Gender | Nationality | E-mail | Career Stage | Role of researcher (in the project) | Reference Identifier | Type of identifier |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

## Role of participating organisation in the project

| | |
|---|---|
| Project management | ☐ |
| Communication, dissemination and engagement | ☒ |
| Provision of research and technology infrastructure | ☒ |
| Co-definition of research and market needs | ☒ |
| Civil society representative | ☐ |
| Policy maker or regulator, incl. standardisation body | ☐ |
| Research performer | ☒ |
| Technology developer | ☒ |
| Testing/validation of approaches and ideas | ☒ |
| Prototyping and demonstration | ☒ |
| IPR management incl. technology transfer | ☐ |
| Public procurer of results | ☐ |
| Private buyer of results | ☐ |
| Finance provider (public or private) | ☐ |
| Education and training | ☐ |
| Contributions from the social sciences or/and the humanities | ☐ |
| Other<br>If yes, please specify: (Maximum number of characters allowed: 50) | ☐ |



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**List of up to 5 publications,** widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

| Type of achievement | Short description (Max 500 characters) |
|---|---|
| *Publication* | *Juan Camilo Valencia Estrada and Jorge García Márquez, "On-axis diffraction-limited bi-conical lenses," J. Opt. Soc. Am. A 39, 107-114 (2022)* |
| *Publication* | *Jorge García-Márquez and Juan Camilo Valencia-Estrada "Freeform compound concentrators for indoor optical wireless communications", Proc. SPIE 11495, Nonimaging Optics: Efficient Design for Illumination and Solar Concentration XVII, 114950J (20 August 2020);* |
| *Publication* | *J. Cosmas, B. Meunier, K. Ali, N. Jawad, M. Salih, Y. Zhang, Z. Hadad, B. Globen, H. Gokmen, S. Malkos, M. Cakan, H. Koumaras, A. Kourtis, C. Sakkas, D. Negru, M. Lacaud, M. Ran, E. Ran, J. Garcia, W. Li, L-K Huang, R. Zetik, K. Cabaj, W. Mazurczyk, X. Zhang, and A. Kapovits, "A 5G Radio-Light SDN Architecture for Wireless and Mobile Network Access in Buildings", in 2018 IEEE 5G World Forum (5GWF), 2018, pp. 135-140.* |
| *Good* | *LiFiMAX Solution for indoor networking. Range of products composed of access points managed by a network controller and USB dongles to connect to the user equipment to provide and indoor optical wireless communication network access. See: https://www.oledcomm.net/products/#point-access* |
| *Good* | *LiFiMAX Chipset. Miniaturized optical transceiver to transmit and receive optical wireless communications signals over a range up to 5 m at a data rate of up to 2Gbps. Very small form factor (smaller than a 1 euro cent coin) to ease integration into user equipment. See: https://www.oledcomm.net/products/#ofe* |

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

| Name of Project or Activity | Short description (Max 500 characters) |
|---|---|
| *SOLERIS (French AID agency)* | *2022-2024: "Solutions LiFi sécurisées durcies": Development of beamsteering solutions with sensing implementation in a context of a for cell-free network deployment for headquarters.* |
| *6G-BRAINS (EU, Horizon 2020)* | *2021-2023 - "Bringing Reinforcement-learning Resource Allocation of Radio Light Network for Massive Connections": EU H2020 project focused on the design of 6G networks for Industry 4.0 with the support of Artificial Intelligence and Reinforcement Learning.* |
| *IoRL (EU, Horizon 2020)* | *2017-2021 - "Internet of Radio-Light" EU H2020 project, focused on 5G-based broadband access and accurate location in intelligent buildings with the joint use of mmW and OWC.* |
| *WORTECS (EU, Horizon 2020)* | *2017-2020 - "Wireless Optical/Radio TErabits Communications (WORTECS)": EU H2020 project, focused on ultra-high data rate wireless communication for virtual reality applications using radio communication, OWC and heterogeneous networking.* |
| *LATCAB (French DGAC agency)* | *2018-2019 - "Light and dAta Transmission for CABin": French Civil Aviation funded project, focused on in-flight entertainment server access through OWC and fiber optics backbone in aircraft cabin.* |



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Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

| Name of infrastructure of equipment | Short description (Max 300 characters) |
|---|---|
| *Electronics and Optics Lab.* | *- PCB techno drill machine*<br>*- 3D printer machines*<br>*- Fast-prototyping automatic pick-and-place*<br>*- Re-flow oven*<br>*- Serigraphy machine*<br>*- Radiofrequency and signal and network analyzer*<br>*- Signal generator*<br>*- Signal analyzer*<br>*- Power RF amplifier*<br>*- Thermic tests equipment for LED and PCB*<br>*- Various licenses for* |



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

## Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

**Minimum process-related requirements (building blocks) for a GEP**

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

Proposal ID **101139292**

Acronym **Opti-6G**

# 3 - Budget

| No. | Name of beneficiary | Country | Role | Personnel costs/€ | Subcontracting costs/€ | Purchase costs - Travel and substistence /€ | Purchase costs - Equipment/€ | Purchase costs - Other goods, works and services/€ | Internally invoiced goods and services/€ (Unit costs-usual accounting practices) | Indirect costs/€ | Total eligible costs | Funding rate | Maximum EU contribution to eligible costs | Requested EU contribution to eligible costs/€ | Max grant amount | Income generated by the action | Financial contributions | Own resources | Total estimated income |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Eurescom-european Institute For | DE | Coordinator | 325 500 | 0 | 20 000 | 0 | 16 000 | 0 | 90 375.00 | 451 875.00 | 100 | 451 875.00 | 406 687.50 | 406 687.50 | 0.00 | 0.00 | 0.00 | 406 687.50 |
| 2 | Brunel University London | UK | Partner | 301 620 | 0 | 25 000 | 50 000 | 11 000 | 0 | 96 905.00 | 484 525.00 | 100 | 484 525.00 | 484 525.00 | 484 525.00 | 0.00 | 0.00 | 0.00 | 484 525.00 |
| 3 | Universite De Versailles Saint-quentin- | FR | Partner | 288 000 | 0 | 25 000 | 50 000 | 26 000 | 0 | 97 250.00 | 486 250.00 | 100 | 486 250.00 | 486 250.00 | 486 250.00 | 0.00 | 0.00 | 0.00 | 486 250.00 |
| 4 | Runel Ngmt Ltd | IL | Partner | 349 600 | 0 | 30 000 | 172 000 | 6 000 | 0 | 139 400.00 | 697 000.00 | 100 | 697 000.00 | 627 300.00 | 627 300.00 | 0.00 | 0.00 | 0.00 | 627 300.00 |
| 5 | Oledcomm Sas | FR | Partner | 452 400 | 0 | 24 000 | 0 | 6 000 | 0 | 120 600.00 | 603 000.00 | 100 | 603 000.00 | 542 700.00 | 542 700.00 | 0.00 | 0.00 | 0.00 | 542 700.00 |
| | | | TOTAL | 1 717 120 | 0 | 124 000 | 272 000 | 65 000 | 0 | 544 530.00 | 2 722 650.00 | | 2 722 650.00 | 2 547 462.50 | 2 547 462.50 | 0.00 | 0.00 | 0.00 | 2 547 462.50 |



This proposal version was submitted by **Ádám KAPOVITS** on **25/04/2023 16:32:25** Brussels Local Time. Issued by the Funding & Tenders Portal Submission System.

# 4 - Ethics & security

## Ethics Issues Table

| 1. Human Embryonic Stem Cells and Human Embryos | | | Page |
|---|---|---|---|
| Does this activity involve Human Embryonic Stem Cells (hESCs)? | ○ Yes | ◉ No | |
| Does this activity involve the use of human embryos? | ○ Yes | ◉ No | |
| **2. Humans** | | | **Page** |
| Does this activity involve human participants? | ◉ Yes | ○ No | 38 |
| Are they volunteers for non medical studies (e.g. social or human sciences research)? | ○ Yes | ◉ No | |
| Are they healthy volunteers for medical studies? | ○ Yes | ◉ No | |
| Are they patients for medical studies? | ○ Yes | ◉ No | |
| Are they potentially vulnerable individuals or groups? | ○ Yes | ◉ No | |
| Are they children/minors? | ○ Yes | ◉ No | |
| Are they other persons unable to give informed consent? | ○ Yes | ◉ No | |
| Does this activity involve interventions (physical also including imaging technology, behavioural treatments, etc.) on the study participants? | ○ Yes | ◉ No | |
| Does this activity involve conducting a clinical study as defined by the Clinical Trial Regulation (EU 536/2014)? (using pharmaceuticals, biologicals, radiopharmaceuticals, or advanced therapy medicinal products) | ○ Yes | ◉ No | |
| **3. Human Cells / Tissues (not covered by section 1)** | | | **Page** |
| Does this activity involve the use of human cells or tissues? | ○ Yes | ◉ No | |
| **4. Personal Data** | | | **Page** |
| Does this activity involve processing of personal data? | ◉ Yes | ○ No | 38 |
| Does it involve the processing of special categories of personal data (e.g.: genetic, biometric and health data, sexual lifestyle, ethnicity, political opinion, religious or philosophical beliefs)? | ○ Yes | ◉ No | |
| Does it involve profiling, systematic monitoring of individuals, or processing of large scale of special categories of data or intrusive methods of data processing (such as, surveillance, geolocation tracking etc.)? | ○ Yes | ◉ No | |
| Does this activity involve further processing of previously collected personal data (including use of preexisting data sets or sources, merging existing data sets)? | ○ Yes | ◉ No | |
| Is it planned to export personal data from the EU to non-EU countries? Specify the type of personal data and countries involved | ○ Yes | ◉ No | |
| Is it planned to import personal data from non-EU countries into the EU or from a non-EU country to another non-EU country? Specify the type of personal data and countries involved | ○ Yes | ◉ No | |
| Does this activity involve the processing of personal data related to criminal convictions or offences? | ○ Yes | ◉ No | |
| **5. Animals** | | | **Page** |
| Does this activity involve animals? | ○ Yes | ◉ No | |
| **6. Non-EU Countries** | | | **Page** |



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Proposal ID **101139292**

Acronym **Opti-6G**

| | | | |
|---|---|---|---|
| Will some of the activities be carried out in non-EU countries? | ◉ Yes | ○ No | 34-39 |

UK and Israel

| | | | |
|---|---|---|---|
| In case non-EU countries are involved, do the activities undertaken in these countries raise potential ethics issues? | ○ Yes | ◉ No | |
| It is planned to use local resources (e.g. animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, etc.)? | ○ Yes | ◉ No | |
| Is it planned to import any material (other than data) from non-EU countries into the EU or from a non-EU country to another non-EU country? For data imports, see section 4. | ○ Yes | ◉ No | |
| Is it planned to export any material (other than data) from the EU to non-EU countries? For data exports, see section 4. | ○ Yes | ◉ No | |
| Does this activity involve low and/or lower middle income countries, (if yes, detail the benefit-sharing actions planned in the self-assessment) | ○ Yes | ◉ No | |
| Could the situation in the country put the individuals taking part in the activity at risk? | ○ Yes | ◉ No | |

| 7. Environment, Health and Safety | | | Page |
|---|---|---|---|
| Does this activity involve the use of substances or processes that may cause harm to the environment, to animals or plants.(during the implementation of the activity or further to the use of the results, as a possible impact) ? | ○ Yes | ◉ No | |
| Does this activity deal with endangered fauna and/or flora / protected areas? | ○ Yes | ◉ No | |
| Does this activity involve the use of substances or processes that may cause harm to humans, including those performing the activity.(during the implementation of the activity or further to the use of the results, as a possible impact) ? | ○ Yes | ◉ No | |

| 8. Artificial Intelligence | | | Page |
|---|---|---|---|
| Does this activity involve the development, deployment and/or use of Artificial Intelligence? (if yes, detail in the self-assessment whether that could raise ethical concerns related to human rights and values and detail how this will be addressed). | ◉ Yes | ○ No | 36 |

| 9. Other Ethics Issues | | | Page |
|---|---|---|---|
| Are there any other ethics issues that should be taken into consideration? | ○ Yes | ◉ No | |

I confirm that I have taken into account all ethics issues above and that, if any ethics issues apply, I will complete the ethics self-assessment as described in the guidelines How to Complete your Ethics Self-Assessment ☒



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# Administrative forms

Proposal ID **101139292**

Acronym **Opti-6G**

## Ethics Self-Assessment

### Ethical dimension of the objectives, methodology and likely impact

REQ Humans: This requirement refers to activities involving work with human beings that are not part of the staff of the participants. In the course of the implementation of the project it may occur that collection of personal data, interviews, observations, tracking or the secondary use of information provided for other purposes, e.g. other projects, officially collected information, social media sites, etc. becomes necessary for the implementation of the project.
REQ Personal Data: This requirement refers to activities that involve collection, processing and storage of personal data. In the course of the implementation of the project it may occur that personal data are collected, processed and stored originating for example from interviews, surveys, questionnaires, direct online retrieval etc. The highest ethics standards and legal requirements will be applied when dealing with such data. Sensitive personal data (religion, sexual orientation, race, ethnicity, etc.) will not be collected. A data protection officer will be appointed.
REQ Non-EU countries: This requirement refers to the activities conducted in relation to partners in non-EU countries and other international cooperation partners. The EU has adopted an adequacy decision on the basis of Article 45 of Regulation (EU) 2016/679 (GDPR) for several countries, among others the U.K., Switzerland, Canada and the Republic of Korea. The effect of such a decision is that personal data can flow from the EU (and Norway, Liechtenstein and Iceland) to that third country without any further safeguard being necessary. Although not members of the EU, Norway, Liechtenstein an Iceland are members of the European Economic Area (EEA). The GDPR was incorporated into the EEA agreement and became applicable in these three countries.
REQ Artificial Intelligence: This requirement refers to the activities involving the development, deployment and use of artificial intelligence (AI)-based systems or techniques. The project will comply with the ethics standards and act in compliance to the Assessment List on Trustworthy Artificial Intelligence proposed by the high-level expert group on AI. The project will adhere to the recommendations for ensuring human agency and oversight, transparency and accountability. Where applicable the recommendations on privacy and data governance, fairness, diversity and non-discrimination, societal and environmental well-being will be followed.

Remaining characters 2548

### Compliance with ethical principles and relevant legislations

The project team is fully aware of the "Ethics and Research Integrity" regulation, which sets out general ethical principles that should be followed in all research activities. The team members receive training by their organisation on the "Ethics and Research Integrity" regulation. The project will consider all ethics issues and describe how they will be dealt with in deliverable D1.1 “Data Management Plan and Ethics Requirements” due at month 6 of the project. Specifically concerning the identified requirements:
- Concerning REQ Humans and REQ Personal Data the project may face issues that are subject to the GDPR rules and which all pertain to obtaining consent from the affected individuals. Staff members of partner organisations have received training on how to treat data falling under the GDPR rules and some of them assume the role of the data officer in their organisations. Previous experience with handling personal data has been very positive and the project has appropriate information sheet templates, consent form templates and other necessary document templates at hand that will be used to comply with the ethics principles in these areas.
- Concerning REQ Non-EU countries, and as stated above, the project does not plan to import from or export data to Israel or the U.K.
- Concerning REQ Artificial Intelligence, the consortium partners are fully aware of the increasing concern pertaining to the transparency and accountability of the AI/ML tools and algorithms. The project work will take measures to test and assess the execution of the algorithms in closed/sandbox environments in order to be able to make judgements and statements about the degree of achievable transparency and accountability, always ensuring human agency and oversight. The project team is fully aware of the "AI Ethics Guidelines for Trustworthy AI" document, which provides more specific guidance for the development and deployment of AI systems. These guidelines were developed by the High-Level Expert Group on Artificial Intelligence, which was established by the EC. The guidelines are based on four key ethical principles: respect for human autonomy, prevention of harm, fairness, and explicability. They also provide a framework for the assessment of AI systems based on these principles, as well as guidance on issues such as data governance, transparency, and accountability.

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# Administrative forms

Proposal ID **101139292**

Acronym **Opti-6G**

## Security issues table

| 1. EU Classified Information (EUCI)[2] | | | Page |
|---|---|---|---|
| Does this activity involve information and/or materials requiring protection against unauthorised disclosure (EUCI)? | ○ Yes | ◉ No | |
| Does this activity involve non-EU countries which need to have access to EUCI? | ○ Yes | ◉ No | |
| **2. Misuse** | | | **Page** |
| Does this activity have the potential for misuse of results? | ○ Yes | ◉ No | |
| **3. Other Security Issues** | | | **Page** |
| Does this activity involve information and/or materials subject to national security restrictions? If yes, please specify: (Maximum number of characters allowed: 1000) | ○ Yes | ◉ No | |
| Are there any other security issues that should be taken into consideration? If yes, please specify: (Maximum number of characters allowed: 1000) | ○ Yes | ◉ No | |

## Security self-assessment

| |
|---|
| Please specify: (Maximum number of characters allowed: 5000) |

Remaining characters 5000

[2]According to the Commission Decision (EU, Euratom) 2015/444 of 13 March 2015 on the security rules for protecting EU classified information, "European Union classified information (EUCI) means any information or material designated by an EU security classification, the unauthorised disclosure of which could cause varying degrees of prejudice to the interests of the European Union or of one or more of the Member States".

[3]Classified background information is information that is already classified by a country and/or international organisation and/or the EU and is going to be used by the project. In this case, the project must have in advance the authorisation from the originator of the classified information, which is the entity (EU institution, EU Member State, third state or international organisation) under whose authority the classified information has been generated.

[4]EU classified foreground information is information (documents/deliverables/materials) planned to be generated by the project and that needs to be protected from unauthorised disclosure. The originator of the EUCI generated by the project is the European Commission.



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# Proposal template Part B: technical description

## OPTICAL 6G CELL FREE NETWORKS

[This document is tagged. Do not delete the tags; they are needed for processing.] #@APP-FORM-HERIAIA@#

| | | | |
|---|---|---|---|
| 1 (Coordinator) | EURESCOM-EUROPEAN INSTITUTE FOR RESEARCH AND STRATEGIC STUDIES IN TELECOMMUNICATIONS GMBH | EUR | DE |
| 2 | BRUNEL UNIVERSITY LONDON | UBRU | UK |
| 3 | UNIVERSITE DE VERSAILLES SAINT-QUENTIN-EN-YVELINES | UVSQ | FR |
| 4 | RUNEL NGMT LTD | REL | IL |
| 5 | OLEDCOMM SAS | OLED | FR |

## 1.0 Excellence #@REL-EVA-RE@#

The OPTI-6G project provides a broadband optical wireless communications (OWC) Vcsels phase array solution that operates in near Infrared (IR) 1550nm wavelength which does not suffer from interference because of the propagation characteristics of EM waves in this part of the spectrum and provides universal broadband coverage within buildings from a cell free network of OWC access points that are pervasively located within buildings. This broadband system will provide 1 Gbps data rates up to 5 m over a field of emission of 25° or 3 Gbps with an angle of 15° up to 5 m and pave the way towards a beamsteered OWC system, which will ultimately produce much wider access angles.

This system will be supplemented with a sub cm location measurement and sensing solution, whose accuracy has already been proven at 3.5 GHz. The performance of near IR OWC solution will be compared with the alternative 3.5 GHz 5G radio solution.

Cell free near IR networks can be configured to implement multi-connectivity with cell free network thereby improving link quality and reliability, whilst obviating the need for building owners to subdivide their non-public mobile building network into cellular areas, and obviating the need for building owners to request permission from MNOs to use their licensed spectrum since the system operates at the optical unlicensed bands and its AI Based Distributed Scheduler manages any interference between inside and outside access. The novelty of using Vcsel phase array is that it prepares the way towards beamsteering the near IR waveform, the novelty of using 1550 nm Vcsels instead of 940 nm Vcsels is that the former will be much less interfered from natural visible light emissions, so will be much more robust.

The challenge that is addressed by OPTI-6G project is to design low cost, light weight, low electronic footprint OWC Remote Units (O-RUs) that can be elegantly integrated into near IR access points and whose receivers can be integrated into a myriad of user equipment such as extended reality headsets, drones, robots, AGVs etc. and compare it with an existing 3.5 GHz 5G cellfree network solution. The main benefit is the availability of broadband communications services greater than 1 Gbits/sec ubiquitously available throughout buildings from pervasively located OWC license free access points.

It has been shown by project partners in the 6G BRAINS project that user equipment terminals can be located to a few cm accuracy using Time Difference of Arrival (TDoA) by analysing OFDM Pseudorandom Noise (PN) sequence in a 5G sub 6GHz carrier frequency localisation system. Currently, existing OWC localisation solutions that use Received Signal Strength (RSS) alone do not achieve better location accuracy than 1cm–3cm, which is not sufficient for AGVs, XR Headsets, Drones, IoT devices and Robots use cases, so this will be enhanced by combining it with Angle of Arrival (AoA) and/or TDoA to explore if a higher accuracy can be obtained. The benefits of these localisation technologies at near IR OWC will be compared with an existing 3.5 GHz localisation solution.

Designing the OWC system to fit into the confined space of an RU requires a Network Function Virtualisation (NFV) solution, whose cloud computers can be variously located remote from the O-RU access points elsewhere in the Home Cell or in the external Cloud network, and a Software Defined Network (SDN) to intelligently manage and route data to the various parts of the radio-light network. Far-reaching benefits of this architecture are that its common OWC building network resources can be more easily shared between MNOs by slicing and that the NFV solution provides an Application Programmers Interface (API), which allows third party service providers to write specialized network applications to manage multi-MNO networks in homes, businesses and public space buildings and environments such as tunnels, train stations and airports.

OPTI-6G project will also significantly benefit 5G MNOs as it will considerably reduce EM interference, which would have otherwise been generated by Home gNBs thereby increasing throughput in the wider 5G mobile network and improving mobile access to users within buildings without incurring interference with the wider 5G mobile network, thereby increasing the value of their customers' buildings.

The net result is that there will be a considerable reduction of transmission power and EMF radiation levels, so the user equipment will potentially consume one tenth energy, which would result in 90% energy savings, a ten times increase in battery lifetime during use in buildings. This will be compared

with a corresponding sub 6 GHz solution. The combined effects of a reduction in delay spread, due to smaller room geometries, the adapted 3GPP 5G approach and the considerable reduction in propagation delays, is expected to result in a reduction in latency to within 1ms. 5G mobile network users will significantly benefit as they will have the choice of a wider range of network services from third party network and home services providers who for example could provide substantially higher bit rates using OWC frequencies at in-door environments, whilst also significantly reducing the level of EM exposure.

## *1.1 Objectives* and Ambition #@PRJ-OBJ-PO@#

### *1.1.1 Measurable Objectives*

The table below describes the research main objectives ,the partners involved and the initial and final TRL level

1. ***Measurable Objectives***

| TECHNOLOGY | DESCRIPTION | PARTNER(S) | CURRENT TRL | TARGET TRL |
|---|---|---|---|---|
| Near IR OWC 6G 1 to 5 Gbps Cell Free Network and UE | Throughput, latency, Energy Efficiency, ICI and EM exposure KPIs will be measured | OLED, UVSQ | TRL2 | TRL4 |
| mm range Localisation and 1 degree orientation accuracy using RSS and AoA in near IR network | Location accuracy KPI will be measured | OLED, UVSQ | TRL2 | TRL4 |
| mm range Localisation accuracy using TDoA in near IR wireless network | Location accuracy KPI will be measured | OLED, UVSQ, REL, UBRU | TRL2 | TRL4 |
| 1 degree orientation accuracy using TDoA near IR wireless network | Orientation accuracy KPI will be measured | REL, UBRU, OLED, UVSQ | TRL2 | TRL4 |

### *1.1.2 Current Art and Beyond the State-of-the-Art*

#### *1.1.2.1 OWC System*

State-of-the- art

Optical wireless communications (OWC), which consist in transmitting data wirelessly by visible, infrared or even ultraviolet light, have a long history of academic and industrial research, which began in the 1960s with the demonstration of free space optics (FSO) terrestrial links over several kilometers. In 1979, Gfeller and Bapst [Gfeller79] established for the first time an OWC-based indoor wireless local area network (WLAN). This application, called from 2009 Light-Fidelity (LiFi), has known since the 1990s many works, first with infrared light [Kahn97] and then, since the emergence of white LEDs in the 2000s, with visible light sources that opened the way to a ubiquitous lighting and communication infrastructure [Komine04]. From an industrial point of view, it was not until the end of the 2010s that the first LiFi standards were actually published and implemented into commercialized products. The ITU-T G.9991 recommendations, published from 2019 by the ITU [ITU19], are now the basis of a majority of LiFi products put on the market, whereas the IEEE 802.11bb task group is currently developing a WiFi-based LiFi standard that should open new markets. At the same time, industrials are

working on the miniaturization of the electronic components, and in particular, of the optical antennas, in order to reduce their size and cost while increasing their performance, which would then facilitate their integration in consumer equipment (e.g. smartphone, tablets etc.). Figure 0 shows for example an optical antenna developed by Oledcomm, embedding a VCSEL infrared source and high-speed PIN photodiodes receivers and that supports 1 Gbps data rates up to 5 m over a field of emission of 25° or 3 Gbps with an angle of 15° up to 5 m.

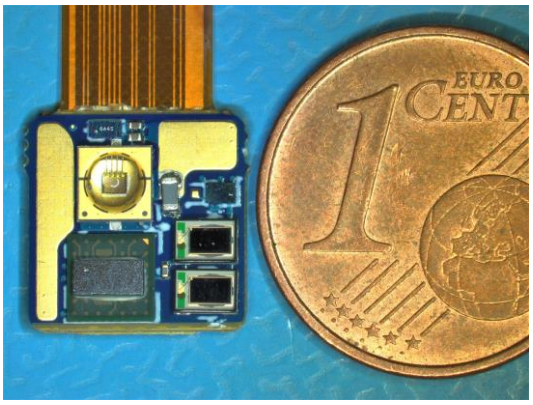



Figure 0: Current optical antenna used in cell-based LiFi systems.

LiFi is therefore an already existing reality, both at the academic and industrial levels. Nevertheless, for the large-scale use of LiFi to be correlated with the deployment of 6G networks, many challenges remain to be solved. In particular, most of the current LiFi products and academic demonstrators rely on the concept of cells, generated by multiple access points (AP) installed at regular intervals on the ceiling to ensure complete coverage of the room and thus continuous connectivity for the user. Such a topology then requires conventional interference management and handover mechanisms to ensure that all user equipment (UE) is continuously connected to the network. Although functional, this cell-based topology has several limitations, similar to that experienced in the RF world. First, it is preferable to have cells large enough to avoid too frequent handovers, which would reduce the time dedicated to traffic transmission in favour of signalling transmissions. At the same time, it is difficult to provide coverage of more than a few square meters with a conventional light source. The signal-to-noise ratio (SNR), and therefore the QoS, degrades rapidly as the receiver moves away from the transmitter's transmission axis, and is therefore generally relatively low at the edge of the cell. In addition, each change of cell results in a short loss of connection, which contributes to further degradation of QoS.
To overcome these problems, cell-free LiFi topologies have been theoretically proved to be a promising solution [ALSatai23].

### Beyond the state-of-the-art

In the OPTI-6G project, we propose to develop an OWC cell-free topology for LiFi applications by developing compact, fast and wide range optical antennas. Such a light source would thus ensure a high SNR link that would be hard to eavesdrop for an attacker who would not be in the line of sight and would thus be more secure than in the usual cell-based topology. A UE could in turn be served by several APs simultaneously, thus making its connection more robust to obstacles. Each AP could also provide several parallel links through wavelength division multiplexing (WDM) to support MIMO modes of operation and thus provide an even greater QoS. The resulting LiFi network could operate alongside RF networks by using similar, if not the same, PHY and MAC layers, generated by a unique remote unit (RU). At the same time, the combination of several Active Optical Units (AOUs) will enable trilateration methods to be used, which will be of great interest to implement optical localization and orientation methods (see Section 1.2.7). Therefore, the proposed network topology would be in line with the overall 6G requirements, as it would be very flexible to adapt to the dynamic user environment and yet provide high QoS and secure service while ensuring sensing at the same time.

### *1.1.2.2 6G Cell Free Network*

Cell Free system comprises a plurality of distributed access points (APs) which simultaneously serve a smaller number of users over the same time/frequency resources based on directly measured channel characteristics. All APs are randomly located in a large area. Furthermore, all APs connect to a central processing unit via a backhaul network, see Figure 1. We assume that all the M APs simultaneously serve all K users using the same time-frequency resource. The APs acquire channel state information through time-division duplex operation and the reception of uplink pilot signals transmitted by the users. The APs perform multiplexing/de-multiplexing through conjugate links on the downlink and matched filtering on the uplink. Closed-form expressions for individual user uplink and downlink throughputs lead to max-min power control algorithms. Max-min power control ensures uniformly good service throughout the area of coverage. A pilot assignment algorithm helps in mitigating the effects of pilots contamination, but power control is far more important in that regard. Cell-Free has considerably improved performance with respect to a conventional small-cell scheme, whereby each user is served by a dedicated AP, in terms of both 95%- likely per-user throughput and immunity to shadow fading spatial correlation. Under uncorrelated shadow fading conditions, the cell-free scheme provides nearly 5-fold improvement in 95%-likelihood per-user throughput over the conventional small-cell scheme, and 10-fold improvement when shadow fading is correlated [NAY17]

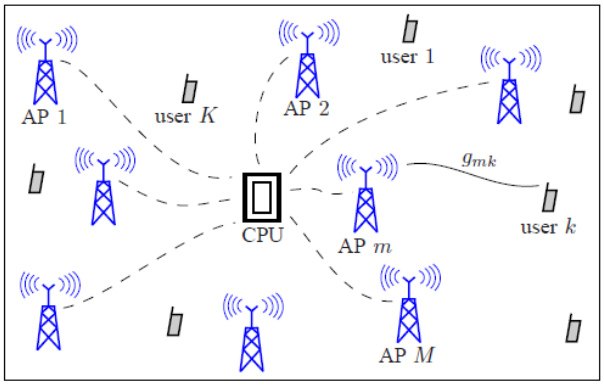



Fig-1 Cell Free Network Architecture

**State-of-the-art**

Despite the clear theoretical advantages that Cell Free systems show in simulations to substantially outperform small-cell systems in terms of throughput and robustness to shadow fading correlation, the reality shows that presently there are no reported commercial deployments of Cell Free networks for the following reasons:

- Implementation complexity related to the need to synchronously combine the signal from the UEs received by multiple APs;
- Implementation complexity related to the need of a Central Network Scheduler that will schedule packets not only in the Time-Frequency domain but also in the Spatial domain due to the fact that multiple APs in different spatial locations are synchronously receiving and transmitting with single UEs in their coverage area;
- Cell Free Networks require a higher front-haul bandwidth compared to regular cellular networks that connect each UE to a single AP;
- The standardization process of Cell free networks is still ongoing in the recent 3GPP releases (Release 18 and 19).

An example of the current art for 5G Private Network architecture is described in Fig 2 below: The system is based on two types of modules, the NIB (Network In a Box) that includes the NgCore the CU and the DU units in one server that is connected through Ethernet over fiber or wireless front-haul to a set of beamforming-capable RUs that wirelessly connect via four dual polarized beams with commercial 5G end devices such as smart phones or any other termination devices. The network achieves a 2 Gbps (0.5 Gbps/beam) aggregate data rate per 100MHz channel in each RU.

The basic architecture includes a 5G RAN PHY Split (ORAN Option 7.2) that enables maximum virtualisation of the base station functions creating a local edge cloud that facilitates the advantages of centralized computing and keeps the fronthaul connection with the edge using limited data overhead and delay (microseconds). The split point protects the data that runs over the fronthaul by taking advantage of the 5G powerful Low Density Parity Check (LDPC) embedded Forward Error Correction

(FEC). The beamforming antenna weights are calculated and optimized in the DU unit but activate the precoding matrix in the RU units along with the (Inverse) Fast Fourier Transforms (FFT/IFFTs) demodulators/modulators and the radio front end. In addition each RU includes a high performance 64 (8x4x2) active hybrid digital/analog weighted beam forming antenna at 3.5GHz (n78 TDD FR1 5G NR band) with 4 dual polarized streams, with low side-lobes and with high antenna gain. Figure 2 illustrates the main network components.

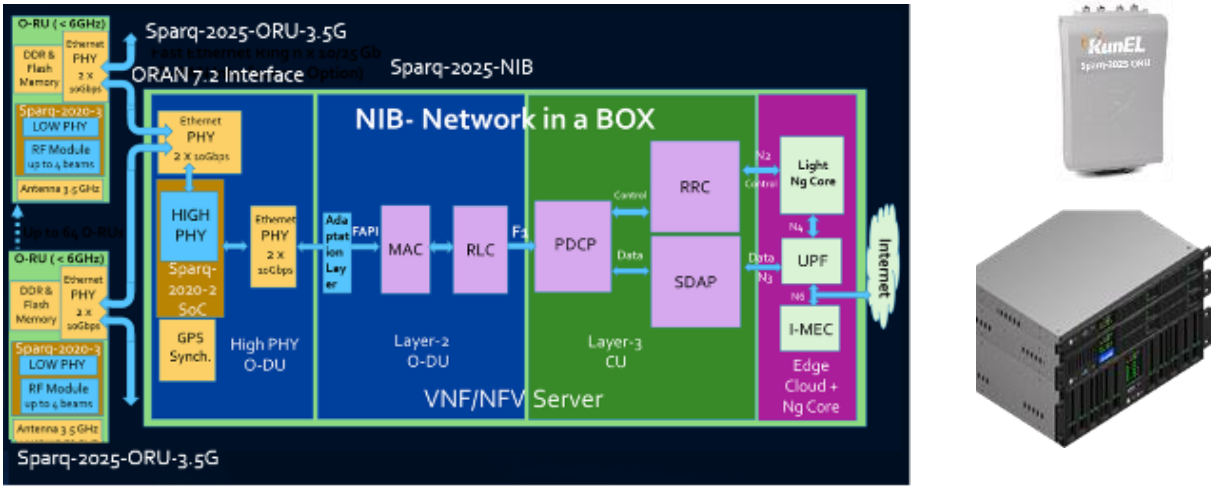



Figure 2 - Left) Private 5G Network functional block diagram. Right) the RU and the NIB hardware

## Beyond the state-of-the-art

In this research the OPTI-6G team will present substantial achievements compared to the current state-of-the-art of Cell-Free Networks including:

- Actual lab implementation of a Cell-Free network at a lab environment with the ability to perform and meet critical KPIs measurements such as throughput, latency, coverage, robustness, UE location accuracy, etc;
- Implementation of the Cell-Free network in near IR optical domain to demonstrate the potential use in indoor applications and services and to evaluate the advantages and disadvantages of such implementations.
- Taking advantage of the Cell-Free network architecture that simultaneously and synchronously connects the UEs to multiple APs to facilitate sub cm accurate UE location measurements.
- Upgrade the DU central scheduler in the Network in the Box (NIB) Module into a disruptive Artificial Intelligence (AI) based cell-free scheduler, responsible of user-centric cluster formation and dynamic optimal packet scheduling in the presence of user mobility, and implementing multi-connectivity using near IR multipoint transmissions. The new cell-free scheduler will centrally control the RUs that will simultaneously communicate with the end devices in the coverage area over the proposed Photonic air interface.

### *1.1.2.3 Localization* and *Orientation in OWC spectral bands*

## State-of-the- art

The main technique of localization with OWC is mainly the use of the received signal strength (RSS), the time of arrival (TOA) or time or phase difference of arrival (TDoA/PDoA) and the angle of arrival (AOA) to extract position information, either by trilateration or by triangulation [ZHY18]. The RSS method is the most popular and algorithms coupled with Kalman filters have been proposed to improve the positioning resolution and responsiveness to motion [ZWZ20]. Also precise round trip (RT) TOA measurements are possible, which are based on several APs and can be implemented using existing fields in the physical layer preamble [KKM21], together with fine timing measurement (FTM) available in the medium access control (MAC) layer of IEEE Std. 802.11 (Wi-Fi).

However, most of these works are based on simulations or experiments in simple or size limited environments [ZHY18], [THA22]. Only few of them address 3D positioning or information about orientation of the receiver with experimental part in large environment. Inclusion of the altitude parameter requires more complex equations and/or additional sensors like inertial sensors or camera [YHV14], [MCP15].

### Beyond the state-of-the-art

In the OPTI-6G project, we propose to aim sub centimetre accuracy in 3D including the information of orientation angle of the target, based on several improvements: a) develop a combined method with RSS and RT TOA, which, to our knowledge, has not been addressed yet; b) combine this first sensing approach with AOA measurements in a different approach than in the literature [THA22] [YHV14] and based on previous work on beam steering [MCP15]. We propose to modulate the angle of emission of the source and, thus, simplify the receiver which is an important point in the case of simple embedded receivers.

#### *1.1.2.4 Localization* and Orien*tation in MW Cellular Bands*

### State-of-the- art

Localization of UEs using Time of Arrival (ToA) from sub-6 GHz wireless and Received Signal Strength (RSS) from Optical Wireless Infrared techniques can be used to obtain an accuracy of less than 1 cm, however it is highly dependent on the continuous direct line of sight access between the gNB access points and the UE [ACZ22]. In 6G BRAINS project, it has been shown in a laboratory experiment that 2 to 3 mm localisation accuracy can be obtained using TDoA in a Sub-6GHz 5G OFDM Network [MKG23] [MES23].

If there is no direct line of sight access to four or more gNB access points then location ambiguity is introduced, and so other techniques can be used to maintain localization such as dead reckoning from Inertial Measurement Units (IMU) or Angle of Arrival (AoA) from received radio signatures. In 6G BRAINS project, it has been shown that 2 - 3 degrees orientation accuracy can in principle be obtained in a simulation experiment using Matlab's Siteviewer 3D radio propagation model [ECS22] [CSB22].

If the received radio signatures estimate Angle of Arrival (AoA) from more than two access points or AoA and distance from one access point, then some form of interim measure for obtaining location can continue to be made. Distance and speed of user equipment can be estimated using Orthogonal Time Frequency Space (OTFS) modulation from one access point due to its operation in the delay and doppler domain as opposed to using Orthogonal Frequency Division Multiplexing (OFDM) which operates in the time and frequency domain.

### Beyond the state-of-the-art

In the OPTI-6G project, we propose to test if more accurate localization of UEs can be obtained using Time of Arrival (ToA) from near IR OWC because of the greater system bandwidth that can be used to lower the levels of noise which are expected.

#§PRJ-OBJ-PO§#

## *1.2 Methodology* #@CON-MET-CM@# #@COM-PLE-CP@#

### *1.2.1 Use Cases*

#### *1.2.1.1 Industrial Use Case - AGVs, XR Headsets, Drones, IoT devices and Robots*

Private 5G/6G and enterprise Wi-Fi both offer high speeds and high device density that are suited to Industry 4.0 (I4.0). Both have similar deployment and maintenance costs, but each is ideally suited to different use cases. Frequently, both will be deployed together in an I4.0 factory, will complement each other with their respective strengths, and can co-exist without interference [Tea22]. 5G/6G NPN offers more benefits in terms of security and guaranteed throughput, low latency, high location accuracy and high reliability performance.

OPTI-6G will remove the barrier of deploying Non Public Networks (NPN) with a zero-touch control, cell free network where multiconnectivity and macrodiversity with multipoint transmissions is implemented, which increases throughput, increases security and reduces interference. People are reluctant to use 5G/6G because it is more difficult to deploy so this project will demonstrate how easily is to deploy 5G/6G and develop 5G/6G-enabled AGVs, XR Headsets, Drones, IoT devices and Robots in particular those that require high location and orientation accuracy, high Data rate, high Coverage distances, fast movement speed of the user, time critical handover support, user equipment density, user equipment size and cost, energy efficiency (user equipment battery lifetime), and service availability. However, Wi-Fi is a complementary interim solution for algorithms and systems development but without security and guaranteed throughput, low latency, high location accuracy and high reliability performance provided by 6G. These attributes could be realized on a 6G network thus providing a substantial differentiation to Wi-Fi networks. Thus, this means that Wireless Computer Communication System development can start using a Wi-Fi solution and subsequently migrate towards a higher performance 6G systems.

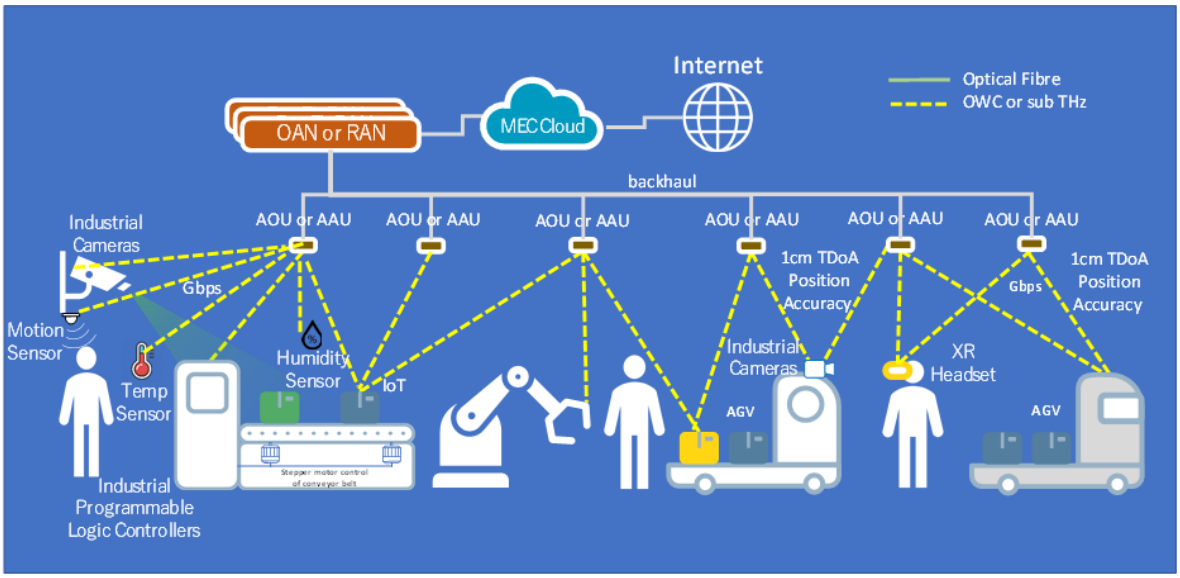



Figure 3: Industrial Use Case

*1.2.1.2 Cell-Free Inside Building Network Access Use Case*Modern buildings are increasingly being constructed with materials that seriously inhibit the propagation of radio signals both within a building and also from outside to inside, such as: foil back plasterboard, insulation board, multi-foil board, flexible foil insulation, low e-glass internal wall insulation, cavity wall insulation, pitched roof insulation and loft insulation for heat insulation purposes. Therefore near-IR OWC access points in each room is an attractive broadband communication solution for the buildings

Building owners have been increasingly turning to the deployment of cellular home networks (HgNBs) in their buildings because they operate in licensed spectrum that can avoid interference and congestion.

Regrettably, these deployments require the permission of Mobile Network Operators (MNOs) due to the regulations and their potential risk of interference with the main transmitted signals from the public mobile network operators (MNO) base stations (gNB). However, MNOs have only had the capacity to analyze their largest customers' deployment requests thereby losing a large market opportunity. Since each building requires to be subdivided into a cellular network of HgNB deployments for each MNO in each room coverage area, this is very costly and inconvenient for the building owner. A cell free network with distributed patch antenna can provide an excellent low cost solution that also eliminates the need for building owners to request permission from MNOs to use their licensed spectrum due to the fact that the proposed solution uses unlicensed optical spectral bands and the AI Based Distributed Scheduler that controls the Cell Free Network takes care also of any interference between inside and outside access and also obviates the need for building owners to subdivide their building into cell areas.

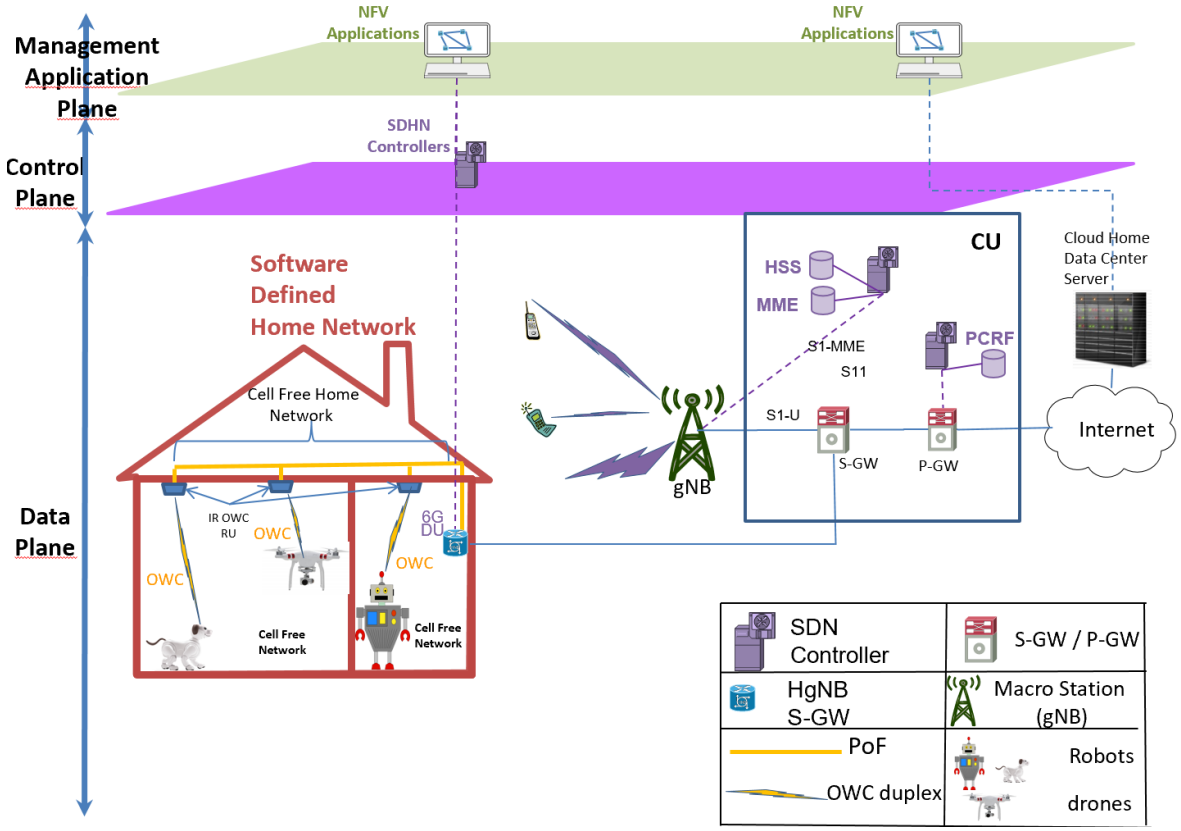



Figure 4: *Cell-Free Inside Building Network Access Use Case*

This technology can also be applied to other indoor environments such as Tube Stations, Underground Pedestrian tunnels etc.

### 1.2.2 6G Communication and Location Sensing System Concepts

There is an expectation for significant KPI improvements in 6G Networks compared to 5G KPIs as can be seen in the table below [EIC22]:

| Key performance indicators (KPI) | 5G | 6G | Improvement factor |
|---|---|---|---|
| Peak data rate (in Gbps) | 10 | 100 to 1000 | 10 to 100 |
| User experienced data rate (in Gbps) | 0.1 | 1 to 10 | 10 to 100 |
| User plane latency (in ms) | 1 | 0.1 | 10 |
| Connection density (in devices/km²) | $10^6$ | $10^7$ to $10^8$ | 10 to 100 |
| Reliability | 99.999% | 99.99999% | 100% |
| Energy efficiency | 1× | 5× to 100× | 5 to 100 |
| Spectral efficiency | 1× | 2× | 2 |
| Positioning (in cm) | 20 to 100 in 2D | 1 in 3D | 20 to 100 |
| Jitter, i.e. latency variations (in µs) | – | 0.1 to 1000 | – |

The aim of the OPTI-6G project is to compare some of the above KPI performances in different 6G architectures (Cellfree vs Cellular) and air interface spectral bands ( sub 6 GHz vs OWS transmissions, Round trip time (sensor to controller to actuator), reliability (packet error rate within latency reqs.), data rate, packet size, covered distance (from an access point), movement speed of the user, time critical handover support, user equipment density, user equipment size and cost, energy efficiency (user equipment battery lifetime), location and orientation detection accuracy, service availability.
The UE positioning accuracy KPI will be investigated in depth using different spectral bands (sub 6 GHz and near IR) and different measurement methods ( based on Time Difference of Arrival (TDoA) or based on Angle of Arrival ( AoA))

#### Optical Wireless

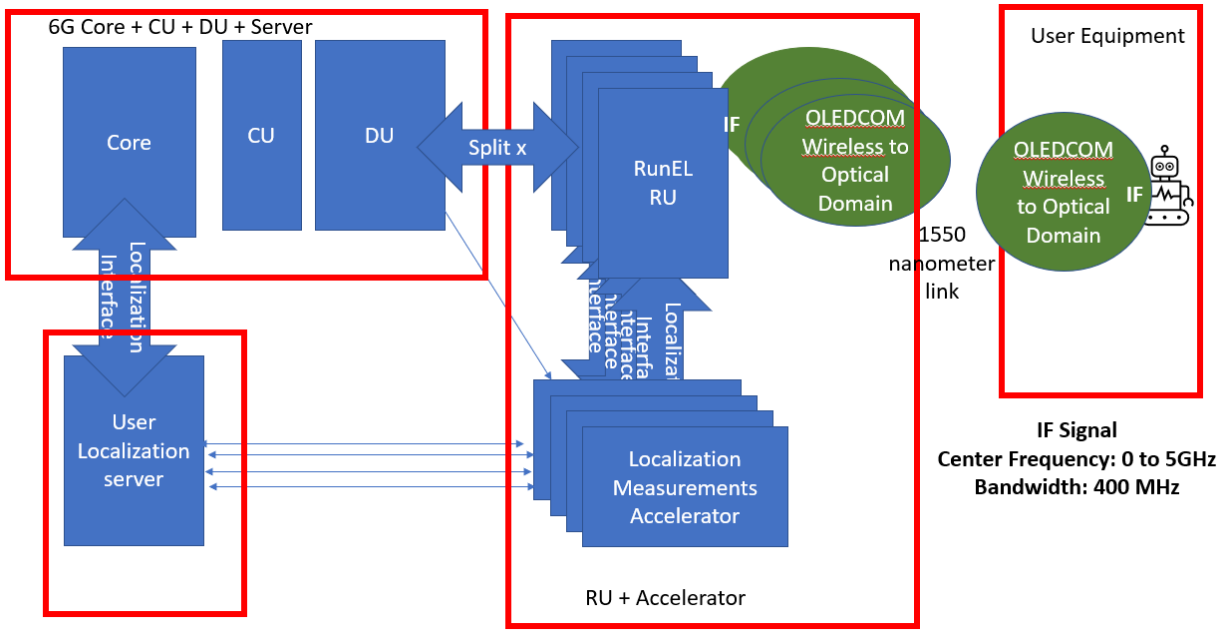



Figure 5: Optical 6G Communication and Location Sensing network

OPTI-6G OWC Communication and Sensing network consists of a complete 6G Core, Central Unit (CU), Distributed Unit (DU), Remote Unit (RU) and an Active Optical Unit (AOU) consisting of an Optical Vcsel Phase Array Emitters/Sensors. OPTI-6G OWC UE is a commercial 5G UE equipped with an AOU interface. The novelty of using Vcsel phase array is that it prepares the way towards beamsteering the near IR

waveform, the novelty of using 1550 nm Vcsels instead of 940 nm Vcsels is that the former will be much less interfered from natural visible light emissions, so will be much more robust.
The near IR uses Vcsel emitter with an angle of emission around presently 25° and with a power that respects eye safety. The source is modulated with classical OFDMA modulations by the RF coming from the RU. The receivers are composed of several silicon photodiodes and the module can operate at distances of several meters with a data rates up to the Gb/s. The pattern of emission on both the AOU and UE sides is known and a position modulation will be added. This modulation can be demodulated at the receiver and the Angle of Arrival as well as the received signal strength (RSS) of the optical signal deduced. Considering several emitter, 3D position can be computed, either on Users with calculus capacity, either up in the core for light users that should just have to communicate the AoA and RSS. At the same time, bidirectional communication between the AOU and UE enables RT TOA/TDoA measurement, so that combination between the three methods (RT TOA/TDoA, AOA and RSS) may eventually provide sub-mm/degree location and orientation accuracy.

### *1.2.3 Underlying Technologies that partners bring to the Integration of Optical and Wireless Technology Solution*

#### *1.2.3.1 Cell-Free Network with AI Based Distributed Scheduler*

The implementation of the Cell-Free Network that will be applied to near-IR OWC networks in the OPTI-6G project is described in Fig 8 below. The Cell-Free Architecture of the Communication Network perfectly matches the requirement for fast, accurate and low latency position measurement of UE units such as AGVs, XR Headsets, Drones, IoT devices and Robots. Since multiple Radio Units (RU) are simultaneously and synchronously communicating with the UEs and accurately measuring the Time Difference of Arrival (TDoA) of signals transmitted by the UE to at least 3 RUs from which the UE 3D location is accurately calculated in real time without the need of hand-offs between the RUs (which is usually required in standard “cell” based networks.

The Cell-Free network consist of two types of modules:

- NIB – Network in a Box
- RU – Radio Units at 3.5GHz band

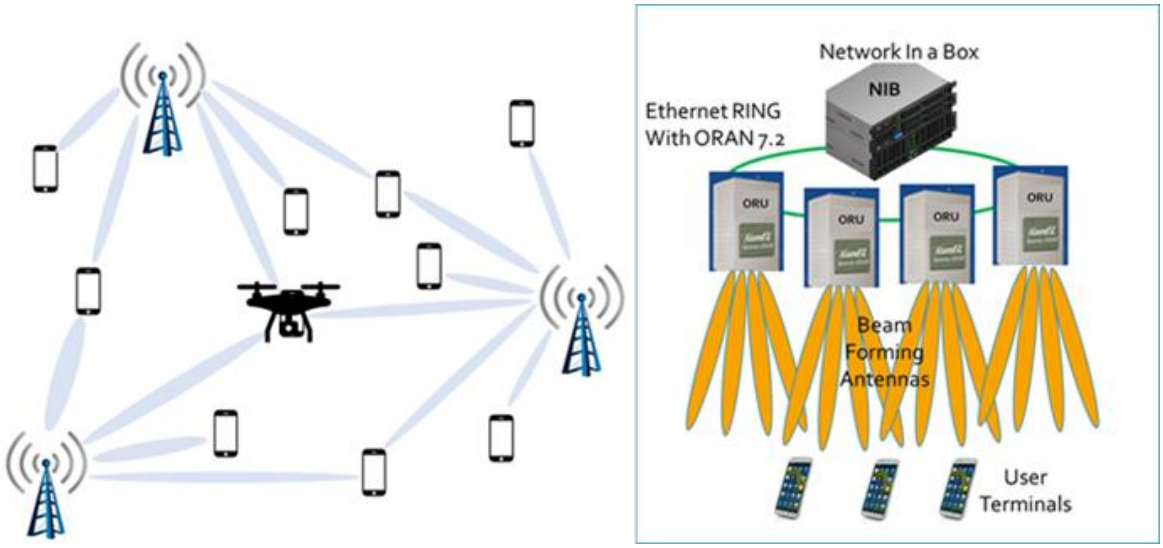



Fig 8: Cell-Free Architecture (Left ) and Testbed used in Cell-Free lab test (Right)

AI Based Cell-Free Scheduler -The RAN MAC Scheduler in Fig 8 is a complex module that schedules the data resources to/from the base stations to/from the UE terminals in an optimal way taking into account multiple variables that can influence the scheduling time/frequency/space resources matrix

such as: Beam selection, Quality of Service/experience, slicing, Modulation scheme, Traffic, Application, Interference, Reliability, User Type, Latency, priority, Buffers status, Optimization target, etc.

This scheduling problem gets even more complex when the Cellular Network becomes a distributed Cell-Free Network. This is because it needs to take care of UEs talking with various Access Points (APs) simultaneously and synchronously.

As a result of the increased Distributed Cell-Free Scheduler complexity and the lack of a feasible model, the traditional ways of scheduler implementation such as Round Robin or Proportional Fair (PF) algorithms cannot deliver an optimal desired solution. Thus, we must look towards various Artificial Intelligence (AI) optimization algorithms (such as metaheuristic, evolutionary, bio-inspired, pareto optimization algorithms, etc...) to search for a feasible optimal solution for this challenging problem.

This research is undergoing in RunEL at the 6G BRAINS Horizon 2020 Project [Hou22] will be extended and applied in the OPTI-6G project with regards to Distributed Massive MIMO near IR Cell-Free Networks. The AI based Scheduler Technology that has been verified in simulations using commercially available Neural Network (NN) components (such as those supported by NVIDIA GPUs), will be validated in a laboratory in OPTI-6G project.

#### *1.2.3.2 Optical Vcsel Phased Array Emitters/Sensors -*

A VCSEL is a vertical cavity surface-emitting laser – a kind of laser diode which emits in a direction perpendicular to the wafer surface – in contrast to the more common edge-emitting semiconductor lasers. Such a laser can produce only quite moderate powers in the milliwatt region. However, it is possible to fabricate two-dimensional arrays containing many such VCSELs on a common semiconductor chip and to utilize the combined emission of those. One can then obtain several watts, dozens or even hundreds of watts, depending on the number of used emitters. Therefore, a 2-D VCSEL array containing many thousand emitters can compete with diode bars and (combining multiple arrays) even with diode stacks based on edge-emitting semiconductor lasers.

For this project we will use vertical cavity surface emitting 1550 nm lasers (Vcsel) (instead of LEDs) to create a Gbps optical transceivers Gbps, and using phase arrays of VCSEL allows a new step to apply some beam steering techniques to make a prototype using beam steering solution (propagation distance 5 to 10 meters) to cover a much higher area

#### *1.2.3.3 Accurate Position and Orientation Estimation*

Estimating Time Difference of Arrival

RunEL has developed a patent pending algorithm that promises to enable accurate measurements of 5G User equipment (UE) 3D location with ultra low levels sub cm resolution resulting in less than 1 cm accuracy. The algorithm is based on accurately measuring the Time of Arrival (ToA) of data packets arriving from the UE to the 5G Open Radio Unit (O-RU) at sub 3.5 GHz frequency. In order to reach the required sub cm resolution, it is required to measure the ToA at accuracies levels of sub picoseconds, which is a real challenge [BGG23]. The 3D accurate location of the 5G UE is calculated by the use of Time Difference of Arrival (TDoA) measurements from at least 3 different locations and then exercising triangulation techniques.

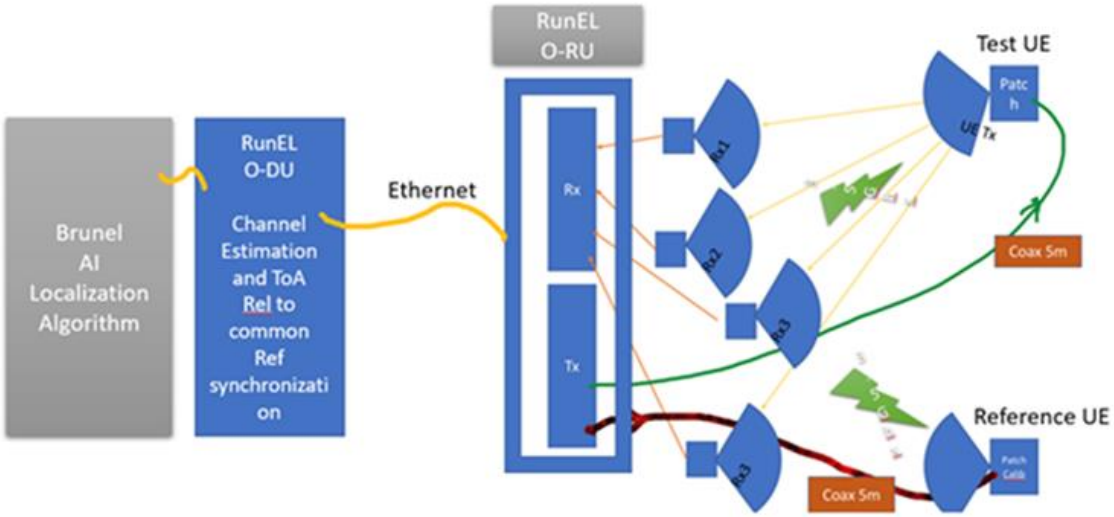



Figure 10: presents the laboratory setup at BREL hat is used to measure the Test UE location.

The setup consists of the REL O-RU and O-DU and the Brunel Localization Measurement Algorithm Application installed on a X86 based server. The 5G OFDMA signals transmitted over the air at 3.5GHz by the Test UE and the Reference UE patch antennas are received by the O-RU Receiver (Rx) via 4 different Rx antennas located in the four corners of ceiling of the tests room. All 4 antennas are connected to same O-RU. The O-RU continuously measures the TDoA of the 5G signals coming from the Test UE and the Reference UE, used for calibration, and transfer the results to the O-DU that stores them on agreed memory sharing register. The Brunel Localization Algorithm reads the ToA registers and calculates the Test UE location.

The challenge in the OPTI-6G project is to show that the patent pending algorithm that promises to enable accurate measurements of 5G User equipment (UE) 3D location with ultra low levels 0.1mm resolution resulting in less than 1 cm accuracy on an OFDM PN sequence at 3.5 GHz, can also measure distance with the same level of accuracy on an OFDM PN sequence at near IR Frequency band.

### Estimating Angle of Arrival frequency

Learning Integrated Reinforcement Learning (DLIRL) algorithm is used in Beyond Fifth Generation (B5G) networks for calculating the Angle of Arrival (AoA). The Smart Base Station (SBS) in B5G networks computes the direction of user equipment (UE) based on the sensing of isotropic transmissions from UEs in coverage area. Further, to improve the Signal to Interference Noise Ratio (SINR), the direction is optimized with reinforcement learning. This includes alternate path finding during obstruction and steer the beam appropriately between SBS and UE. In the proposed Deep Learning Integrated Reinforcement Learning (DLIRL), supervised learning for Deep Neural Network (DNN) and deep Q learning scheme for the deep reinforcement learning based neural networks are realized using a digital twin environment. The algorithm comprises of online learning phase for training the weights and working phase for carrying out prediction. Simulation results, using Matlab's Siteviewer 3D radio propagation model, confirm the efficacy of the proposed DLIRL in direction finding as the spectral efficiency and BER of the B5G system are improved as compared to its counterparts in terms of spectral efficiency (11 bps/Hz at SNR=10 dB) and BER performance ($10^{-5}$ at SNR= 10 dB). The DLIRL system has Angle accuracy towards UE location with a deviation of ±2°, whereas RL has a deviation of ±3° and DNN's deviation is ±5° [ECS22]. .

These experiments will be performed in 6G lab shown in Fig 11, which consists of a (L=2.5m, W=2.5m, H=2.2m) aluminium frame on which DU and four RUs can be located without damaging the room and a flat bed plotter on the floor on which a moving user equipment can be accurately moved with mm accuracy up/down and left/right under computer control.

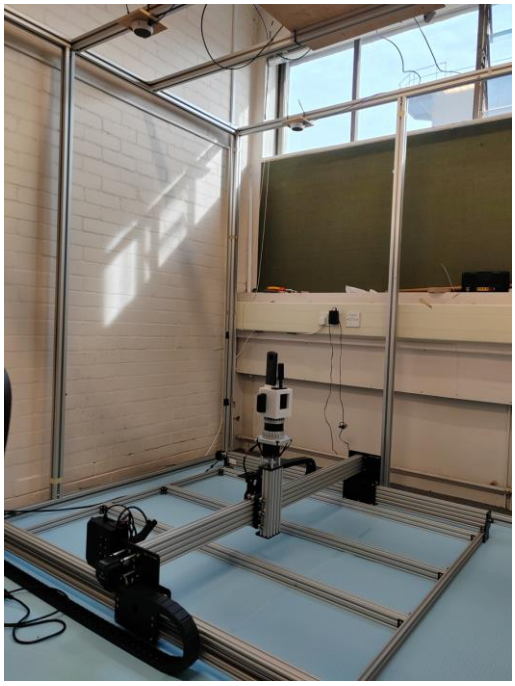


Fig 11: Brunel 5G/6G Lab

For OWC with near IR frequency, in addition to the TDoA method just described, location and orientation estimation is also possible with Angle of Arrival, which are possible to be measured with a single detector and combined with inertial sensors or camera [YHV14][MCP15]. But this addition is not really compatible with simplified embedded user equipment. The new AOA setup proposed by UVSQ is based on the angle modulation of an IR source with known beam pattern, enabled by the steering mechanism, and measurement of intensity on the receiver. Angle of Arrival from several sources can be estimated and, similarly to TDoA, an algorithm can then calculate the 3D position using a triangulation process. Orientation of the photo-receiver can also be measured. At the same time, RSS from several sources can be estimated and used to provide an additional location estimation, which can eventually increase the accuracy through fusion. The target accuracy is therefore within the centimeter.

The challenge in the OPTI-6G project is to show that the predictions obtained from simulation model can be realised in practice for a system operating in near IR frequency.

### *1.2.4 International research and innovation activities whose results will feed into the project*

| Project Title | Project Aims | Project Outcomes |
|---|---|---|
| IoRL (Internet of Radio Light) H2020-761992 IoRL H2020-ICT-2016-2, July 2017-Dec 2020 Internet of Radio Light – IoRL (5g-ppp.eu) | The Internet of Radio-Light (IoRL) project develops a safer, more secure, customizable and intelligent building network that reliably delivers increased throughput (greater than 10Gbps) from access points pervasively located within buildings, whilst minimizing interference and harmful EM exposure and providing location accuracy of less than 10 cm. | Outcomes of the project will be used in WP 4, namely: Shared interest: Openstack MEC building. Localization and intelligent services VNFs and their end user applications. Planning of demonstrations. Concrete collaboration: Localization and intelligent services VNFs and their end user applications. Planning of demonstrations. AI enhanced VNFs for network and drone management. |
| 6G BRAINS (Bringing Reinforcement learning Into Radio Light Network | 6G BRAINS aims to bring AI-driven multi-agent Deep Reinforcement learning (DRL) to perform resource | Outcomes of the project will be used in WP 4 and WP 5, namely: Shared Interest: |

| | | |
|---|---|---|
| for Massive Connections) Grant Agreement number: H2020-101017226, Jan 2021 – Dec 2023 6G BRAINS – 6G BRAINS (6g-brains.eu) | allocation over and beyond massive machine-type communications with new spectrum links including THz and optical wireless communications (OWC) to enhance the performance with regard to capacity, reliability and latency for future industrial networks. | 1 cm location Accuracy of 5G enabled sensors, cell free network |
| SNS Stream A 01-01 Green Radio Technology-BeGreen | Development of Technologies for Energy Efficient Networks such as AI Based Digital Predistortion, and Envelope Tracking , Energy Optimization XApps and Slices, etc. | Low energy consumption, and Low electromagnetic emission Radios |
| OLED | | |
| SafeLiFi: Steerable Light Antenna for Enhanced QoS Cell-Free LiFi Networks ANR 2021-2024 (ANR: Frenc National Agency of Reserach) | Development of beamsteering: simulation and modelling of the parameters, design of steerable antenna and new method for sensing of the receiver position and orientation. | Outcomes of the project will be used in WP4 for sensing algorithm and optimization and in WP2 for the design of optical antenna. |
| | | |

### *1.2.5 Interdisciplinary approach*

The proposed research is inherently interdisciplinary since it combines software, hardware, as well as networking implementation, simulation and modelling, which are essential for the realization of the envisioned prototype. It also involves the derivation of signal processing methods which are the outcomes of an intersection of signal processing, communications, and machine learning (this is particularly true in the context of UE devices such as AGVs, XR Headsets, Drones, IoT devices and Robots). Most importantly, the potential of the proposed research lies in the synergy of joint expertise of Cell Free Networking, near IR OWC RSS/AoA/TDoA localisation to produce enhanced throughput, lower latency, higher localization accuracy and increased reliability capabilities that 6G communication provides.

### *1.2.6 Gender Dimension*

The University which is part of the OPTI-6G consortium (UBRU) that has won Athena Swan Bronze award as an evidence of good practice in gender equality within the institution, will advance the representation of women in science, technology, engineering, maths and medicine (STEMM) best practice to the other universities participating in the project (UVSQ). The Athena Swan Charter was established to advance the representation of women in STEM employment in higher education and research. Brunel University's Equality (including gender), Diversity and Inclusion Strategy is found here: https://students.brunel.ac.uk/documents/Policies/equality-diversity-and-inclusion-strategy-social-justice-for-all-2021-2024.pdf. All other OPTI-6G team members have developed and executed their Gender Equality Plan ( GEP)

### *1.2.7 Open Science Practices*

The EC set off the ambition of the European Open Science Cloud: a trusted environment for hosting and processing research data to support EU science in its leading global role. Resonating with the principle of 'as open as possible', OPTI-6G team members fully support and implement the goals of the Open Science Policy under Horizon Europe and thus, appropriate open science practices will be implemented as an integral part of its proposed methodology as described below.

The open science methods that existing partners exercise in their institutions, according to guidelines provided by [htt22], will be adopted in the project and discussed in the D1.1 Ethics and Data Management Action Plan Report, which will be drafted by the consortium at the start of the project in M06.

Open Methods - documenting and explaining all methods and workflows used in scientific research. Most universities recognize that a cornerstone to research integrity is adherence to the principles of open science and responsible research assessment. Many EU universities have signed up to DORA and support the work of the 'UK Forum for Responsible Research Metrics'. Internal monitoring and measurement of research performance is aligned to recommendations produced by the Responsible Metrics Forum and applied sensitively by considering disciplinary norms, staff circumstances and the individuals' protected characteristics. The use of journal rankings or impact factors in the assessment of research quality is not allowed.

Open Access to Publications and Data – making data and publications accessible to the community, which many universities across Europe have invested in heavily in their research information management systems, thereby, making Europe an open access and open data sector leader. All research conducted in European Universities is reported in their Research Archive Database, which is the universities' research information database that aims to provide a comprehensive record of publications and other research outputs and activities. European universities additionally support an institutional repository which showcases publications and other outputs freely available to the public via the Internet, subject to copyright permissions and publisher policy. Open Access Policy requires that, subject to publishers' copyright permissions, all new research publications and all PhD theses should be deposited into internal repositories. Academic staff authors use their research archive databases to maintain their professional profile and records of activities and outputs, and upload manuscripts accepted for publication to publicly available database. The Research Archive Database also helps university staff to showcase their research in their web profiles on their universities' website (and google scholar, researchgate or scopus), is used to support their universities' submission to any Research Excellence Framework (REF) (particularly for UK HEIs) and reports to funding bodies on research outputs and open access compliance. This is supported by training and on-line guides for staff and students about how to select and prepare data for publication. Additionally, OPTI-6G could share captured and curated data in the open and cutting edge GAIA-X data infrastructure.

Open research – ensuring research is open at every stage of the research process, from working on an original idea, through carrying out the research, to sharing and disseminating the completed research outputs. It is based on the principle that knowledge produces the greatest benefits if it is shared as widely as possible and aims to make the outputs of research freely accessible and usable, and thereby, maximize the possibility of public benefit. The principles of Open Research are reflected in the policies of many public funders and research organisations that promote greater public access to research, and in evolving models of scholarly communication. Change is also driven by the needs of academic communities and stakeholders, among the general public, in industry, and in developing nations.

Open Code - ensuring algorithms and code used in research are human-readable and available. Clearly this is dependent on how much IPR code and algorithms have been introduced as background into the project by each partner, which is usually identified in the Consortium Agreement that is normally based on the Model Consortium Agreement provided by the EU program.

All project researchers are encouraged to use Github (or similar) to share code under appropriate Collective Commons (CC) licence (and Zenodo to assign a DOI for archival purposes). Datasets reported in research papers that refer to data produced in experimental work, are also frequently made available to the wider scientific community through Zenodo research outputs data.

Open Infrastructure - establishing scientific infrastructures that facilitate collaboration use of open infrastructure that facilitates collaboration, including access to our High-Performance Computing (HPC)

facilities, access to our experimental equipment through Teamviewer or X2go, and access to conferring amongst researchers through Teams, Zoom etc… for collaborative meetings.

Open Education - enabling access to OPTI-6G research outputs that would be transformed into teaching and learning materials and postgraduate research topics (e.g. captured datasets, AI and Machine Learning algorithms, and experimental procedures, etc...). Brunel University has begun an open scholarship (sometimes called open education) initiative to ensure that more of our students have access to open resources to support their study.

### *1.2.8 Research Data Management*

Project data including research outputs will be managed in line with FAIR principles (Findable, Accessible, Interoperable, Reusable) (FAIR Principles - GO FAIR (go-fair.org)). A project database will be formed and be controlled by the project coordinators (EUR). This will be a cloud based secured file storage and sharing capability accessed from any device with the right credentials and access rights, available to all consortium partners. Standards and formats associated with storing research outputs will be agreed among the consortium partners during the project initiation phase. Project partners, led by the project coordinators will adopt the usage of the Only Office solution. This enables all members to access files and other collaterals such as contact details, contracts, project plans, newsletters, and financial information in a well managed, controlled and secured manner.

In addition to the above, the project will adopt a research tracker that will hold details of any significant activity such as publications or conferences by a consortium member. The tracker will be accessible to all partners and will be proactively shared by the project coordinators. Regular form of communications will be emails, video meetings and direct calls between the partners as and when needed. All formal project meeting minutes and actions will be documented by the coordinators and be made available to all partners via the Only Office file storage.

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## 2 Impact #@IMP-ACT-IA@#

### *2.1 Pathways towards impact*

(1) Integration of Optical and Wireless Technologies of advanced light related technologies such as LEDs (light-emitting diodes), lasers, outdoor point-to-point devices (FSO — Free Space Optics), point-to-multipoint commercial applications (Li Fi — Light Fidelity) or between devices (OCC – Optical Camera Communication) and Fiber Wireless (Fi-Wi), for the design of novel communication schemes, system architectures and protocols, in order to fully integrate these technologies in the communication infrastructure.

The OPTI-6G project develops

1. A Cell-free network whose signalling load for handoff between cells is no longer increasing with an increasing number of UEs, which means that connection densities of up to one million mobile devices per square kilometre can be supported for mobile mMTC devices.
2. Joint communication and location sensing system that can discern distances to UEs to a few mm accuracy and orientation to a few degree accuracy.
3. OTFS or OTFS-like system which can communicate data to highly mobile devices without suffering from ICI (Inter Carrier Interference) which occurs in OFDM system when sub-carriers lose orthogonality as a result of Doppler frequency changes and that distance and speed of mobile device can be simultaneously estimated.

The impact is that it:

1. Enables additional AOUs to be added without any need for network planning;
2. Reduces the likelihood of NLoS access to RANs/OANs occurring as a result of occluding objects;
3. Overcomes bit errors that occur due to Doppler in fast moving OWC channels.
4. Enables Indoor Cellular Operation at unlicensed Optical Spectral Bands and substantial Capital and Operational Expenditures ( CAPEX and OPEX)

5. Protects the Environment by its substantial reduction in Microwave Radiation and Power Consumption

(2) Hybrid RF/Free Space Optics communications and sensing techniques to underpin selected verticals and applications are in scope.

It has been shown by project partners that the 6G mobile networks can use its sub 6 GHz communication resources to accurately locate the position of user devices to 2 - 3mm using Time Difference of Arrival (TDoA) in a physical experiment, and orientation to 2– 3 degree accuracy using phase of arrival at an antenna array in a simulation experiment.
Since the location estimation algorithm uses a pseudo random noise pattern on OFDM frames to estimate propagation delays of sub 6 GHz frequency transmissions to sub picosecond resolution (equivalent to sub mm resolution), this algorithm can also be applied to estimate propagation delays of near IR frequency transmissions to 0.3ps resolution.
Thus the OPTI-6G location sensing system will support the localisation and orientation accuracy requirements for AR Headsets of 1mm every 100ms, and for collaborating drones of 1mm with cycle time of motion control function: 1 ms, synchronicity 10 ns, whilst both require the provision of localisation with 99.99% reliability.
The impact of this is that a low cost, low energy communication and location sensing system can be used for AGVs, XR Headsets, Drones, IoT devices and Robots equipment which have limited load bearing and power supply capabilities thus allowing them to operate for longer periods of time.
The impact is that it:

1. Accurately measures the location of user devices to 2 - 3mm using Time Difference of Arrival (TDoA), and orientation to 2– 3 degree accuracy using phase of arrival at an near IR sensor array;
2. Uses less power measuring distance and orientation thus prolonging battery life capabilities;

The exact measure of improvement of these impacts will be produced by technical experimentations in the project.

(3) Ultra-low energy and ultra-high capacity solutions for access or end to end connectivity based on optical technologies and their integration within a wireless-optical connectivity continuum.

The OPTI-6G communication system transmits near IR to multiple UEs thereby producing an ultra-high capacity cell free network solution, whilst the localisation and sensing system is used to locate the UEs in their environment to mm location ad degree orientation accuracy.
QoS in this type of EM environment is now dependent on the number of access points that could be scheduled to support the communication between multiple spatially different transmitters and the intended receiver.
The impact is that it:

1. supports larger coverage areas;
2. Uses less power thus can prolong battery life capabilities;
3. Suffers less ICI thus produces a greater goodput;

The exact measure of improvement of these impacts will be produced by technical experimentations in the project.

(4) Our ambition in the OPTI-6G project is to design, develop and exhibit the operation of an Radio Optical 6G, so that it can easily be installed in a wide range of different types of properties and where third party application developers can develop and exhibit their

innovative network services in homes, businesses and public space buildings using open source development environment.

Example services will be developed in the project such as one that ameliorates the need for using HeNB (Home Network Base Stations) or WLAN ( Wireless Local Area Networks) in buildings and eliminates the uplink interference that they induce. This will have the effect of incentivising all building construction companies to install an cell free network throughout their new build developments. The OPTI-6G project will be the first project to integrate cell free sub-THz and near IR 6G network architectures.

One of the growing problems in buildings is interference between the range of different wireless systems in the home, namely: electronic equipment such as microwave ovens, cordless phones, wireless headsets, Zigbee, ZWave, Bluetooth devices, surveillance cameras and other wireless radio networks. Thus the main motivation for buildings has been to switch to home networks that use regulated spectrum such as 6G near IR or sub-THz RUs to avoid interference problems.

However many modern buildings are built with thermal metal clad insulation that severely restricts the propagation of EM waves and with metallized windows, which restrict the propagation of EM waves within and to/from outside the building. In the extreme, some buildings are made from steel shipping containers such as the Starbucks building in Tukwila, Washington USA [http-containers]. Therefore the introduction of modern building materials is making it increasingly difficult for the radio signal from wireless transceivers to provide sufficient coverage inside buildings and so many organizations are attempting to deploy Mobile Network RUs on their premises.

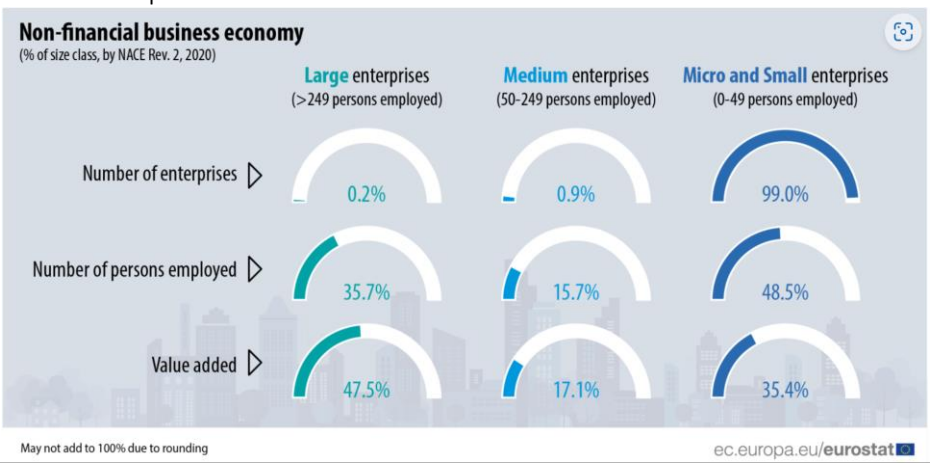



Figure 11: Source Eurostat - Structural business statistics overview - Statistics Explained (europa.eu), 2022

The deployment of HgNB small cells requires the permission of MNOs, due to their potential to interfere with the main gNB transmitted signal, but the MNOs only have the resources to approve installation for their larger business clients (with 50 employees and above) and neglect their smaller clients, who are by far the largest group in the market place, as shown in Figure 11. These smaller clients then become frustrated and return to using WiFi! This trend no longer needs to be experienced worldwide, since a cell free network now removes the need for building owners to request permission from MNOs to use their licensed spectrum since it uses the unlicensed OWC Band and AI Based Distributed Scheduler manages any interference between inside and outside access.

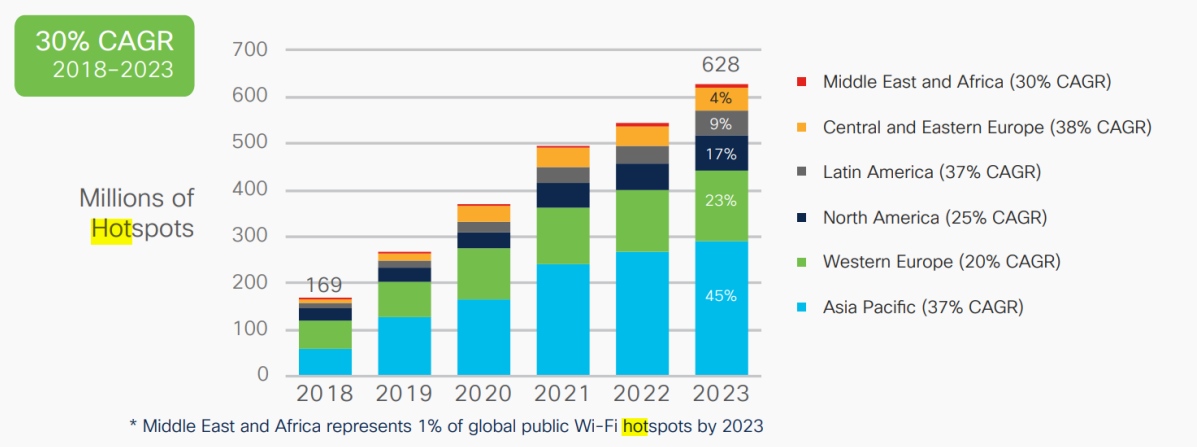



Figure 12: Cisco Forecasts Millions of Hotspots [CIS23]

Furthermore larger clients, who wish to use HeNB small cells in their premises, require separate infrastructure for every MNO that is providing coverage in a building, which is very costly and inconvenient to the client. This has led to the concept of the sharing of mobile network resources, such as with Multi Operator Radio Access Network (MORAN) HeNB systems, and the concept of improving the coverage within buildings such as with Distributed Antenna Systems (DAS), both of which continue to be costly options for the client.
The status quo between mobile network providers and wireless network providers has been that the former have operated their networks in licensed spectrum whereas the latter in unlicensed spectrum. However recently this status quo has been disturbed by the concept of LTE-Unlicensed, License Assisted Access (LAA) and LTE Supplemental DownLink (SDL), which enables network operators the use of the 4G carrier aggregated LTE radio communications in unlicensed spectrum used by Wi-Fi equipment as well as licensed spectrum.

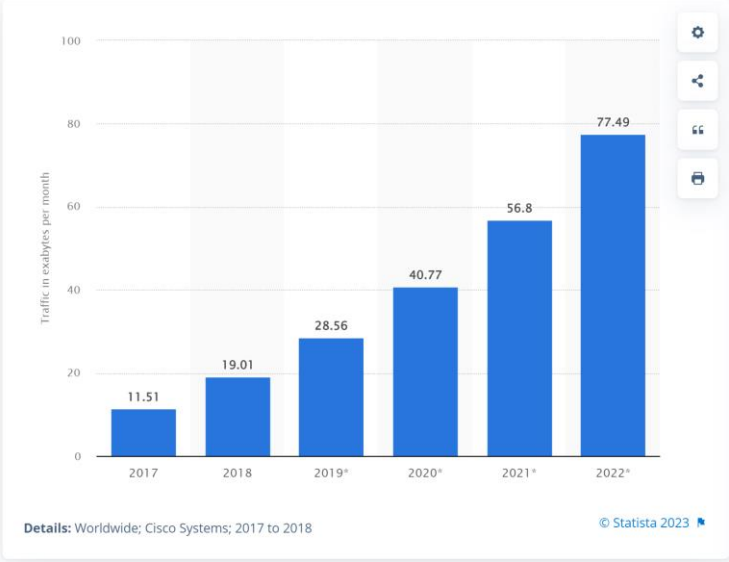



Figure 13: Cisco Forecasts Traffic in Exabytes per month [CIS23]

MuLTEfire has been proposed by Qualcomm, which uses LTE in unlicensed spectrum that does not require an anchor in licensed spectrum, so opening up LTE to organisations that do not own licensed spectrum like ISPs and stadia or conference venue owners. Furthermore the Spectrum Access System (SAS) has been proposed by Google, which uses both 802.11 WiFi and/or LTE in unlicensed 3.5 GHz spectrum for Internet of Things (IoT) devices, which need low

latency, high-performance wireless networks typically over a very short range that can handle up to ten million devices under management.
Clearly with a compound annual growth rate (CAGR) of 46.4% with recorded of 77.49 Exabytes mobile data traffic in 2022, as shown in Figure 13, the competition for spectrum, which is already congested, will only intensify. However the adopting a cell free Mobile Network solutions in near IR frequencies will alleviate the competition for spectrum and accelerate the continued growth of hot spots whose current compound annual growth rate (CAGR) is 30% with 628 Million Hotspots recorded in 2022, as shown in Figure 12,

(5) Overarching impact: 40% of the world communication infrastructure market for EU head quartered companies;

Outside to in-door mobile solutions are trouble free for the consumer to set up but do not work for all locations inside a building. Therefore, indoor mobile access service is of increasing importance to consumers, since 80% of mobile traffic is currently consumed indoors from a combination of traffic from home and office locations, but mobile service from gNB macrocells alone fall short since this dedicated in-building solution to indoor mobile service is complicated by the range of buildings and users to be covered.
Mobile data growth is forecast to have a 53% compound annual growth rate from 2015 to 2020 [Cis16] and 80% of this is generated indoors and is growing 20% faster each year than outdoor wireless traffic [Amd15] [Com16]. However fewer than 2% of commercial and public buildings are currently covered by dedicated wireless indoor solutions [ABI14], therefore the introduction of a commercial and public buildings network solution for mobile networks could have an enormous impact on this market.

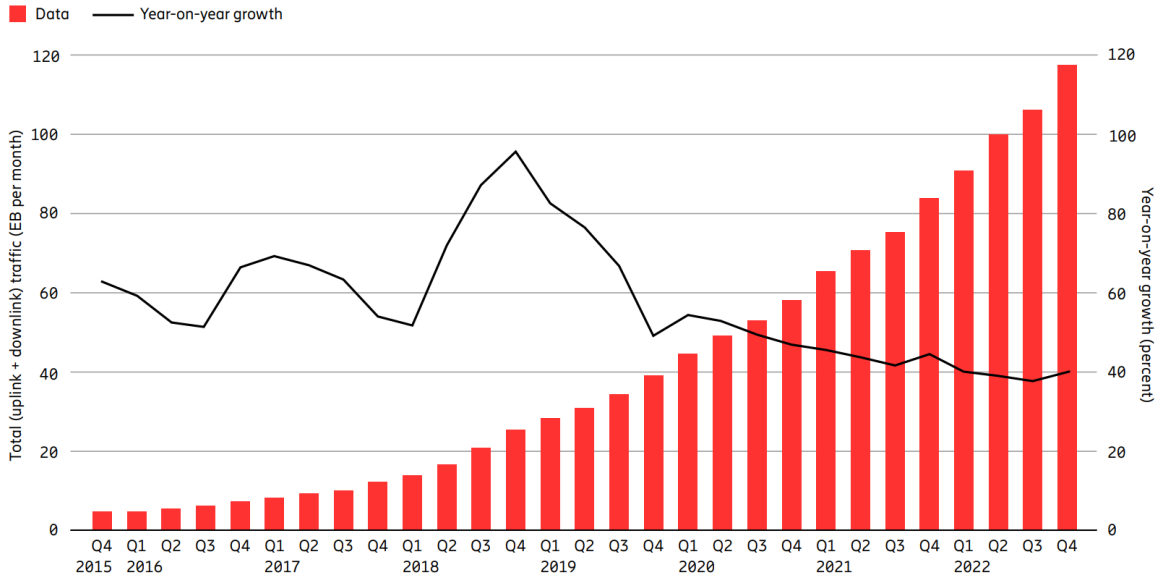



**Figure 14:** Global mobile network data traffic & year-on-year growth (EB per month) **[Eri22]**

Although there has been ever increasing capacity demand for data traffic from Smartphones, mobile PCs and Tablets with voice traffic occupying only a fraction of the total capacity demand, see Figure 14, the gap between traffic increase and revenue growth has been widening, see Figure 15. As a consequence Mobile Network Operators (MNOs) do not have the budget or a valid business case to cover the remaining 98% of pubic/commercial buildings, despite still remaining key stakeholders as they are the spectrum holders/owners. MNOs will

not finance a building access network solution for public or commercial buildings except if there is exceptional number of end users such as in airports and train stations, which typically requires a multi-operator solution, or for a substantial corporate client, which typically requires a single operator solution.

However landlords of buildings need to incorporate mobile services in their buildings as it impacts their ability to let/sell/use their properties and this awareness has led to a change in landlords' and developers' attitudes thereby setting budget aside for providing building access network. Therefore in the absence of an MNO led solution it is expected that there will be landlord owned and third party neutral host building access network solutions occurring.

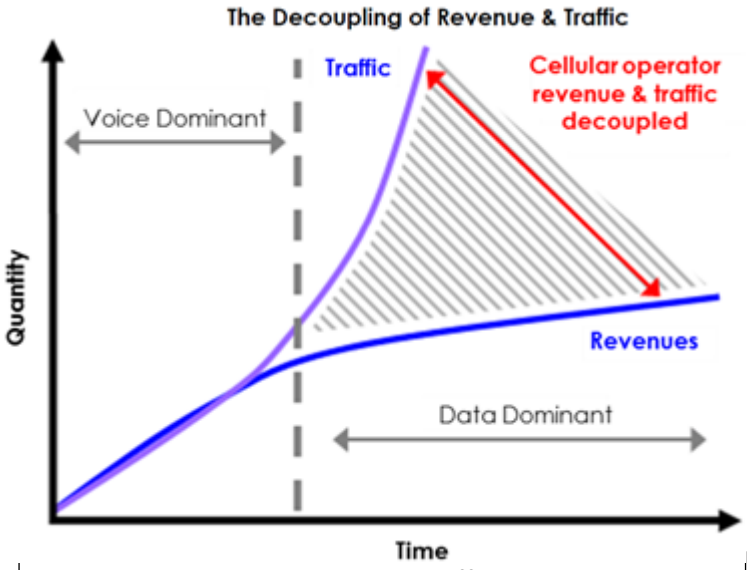



Figure 15: Mobile Revenue and Traffic Decoupling

As explained in Impact 4 above, wireless LAN solutions suffer from interference due to congestion and from the increasingly challenging radio propagation environments that exist in buildings due to the use of materials required for increasingly thermal efficient buildings. Small cell HeNodeB require permission from MNOs to deploy due to their potential to interfere with the main eNB transmitted signal. Distributed Antenna Systems (DAS) need base stations and require sophisticated radio propagation analysis skills for deployment and expensive CAPEX & OPEX. The OPTI-6G project's near IR OWCcell free networks could provide an elegant solution, which could potentially cover the remaining 98% of pubic/commercial buildings for landlord or 3rd party neutral host building network service operators, since permission for communication regulator license for use of spectrum will no longer be needed since interference between inside and outside access points will instead be managed by AI Based Distributed Scheduler.

## (6) Definition of 5G network architecture and of core technological components

The OPTI-6G proposal has been inspired by the 5G PPP Architecture Working Group's report on 6G architecture [MKG23] [MES23] and will contribute toward this group on technical and business aspects of the building infrastructure "vertical market" area with insights and innovations for unlocking network value [Cha13] that take into account all the type of residents that exist within a wide range of different types of properties to provide a holistic solutions for the processing and transport of information within buildings. The 6G network architecture consists of four planes, namely: Management Plane, Control Plane Network Functions and User Plane Network Functions and Infrastructure and Environment Plane across which the sensing stratum spans [MKG23] [MES23]. The OPTI-6G project will focus on developing near IR location sensing system for buildings. It will then show how a location sensing API can be used to develop example Applications and Business Services built from these Localisation Services in Buildings.

The OPTI-6G project will contribute to the definition of 6G network architecture and of core technological components that are required within buildings.

(7) Proactive contribution to the 3G PP standardisation activity on 5G, and to other standardisation activities, e.g. ONF, ETSI-NFV, IEEE; proactive contribution to the WRC 19 preparation for 5G spectrum.

Please see section 2.2.2

## *2.2 Measures to maximize impact* #@COM-DIS-VIS-CDV@#

### *2.2.1 Exploitation*

#### *2.2.1.1 Individual partner exploitation plans*

| No | Partner | Exploitation Plans |
|---|---|---|
| 1 | EUR | EUR will leverage Opti-6G advances primarily in the context of its activities in the SNS JU and 6G IA, but also in nationally funded German research activities targeting 6G, as well as within the Eureka cluster Celtic-Next. Opti-6G results and insights will be used in advising Eurescom's MNO shareholders. |
| 2 | UBru | Brunel will enhance its wireless network facilities with 6G Cell Free sub 6GHZ and Near IR NPNs specifically to measure their accuracy of localisation and to develop high performance Industry 4.0 applications such as those being promoted in this project and promote MSc and BEng student projects to explore merits and performance of 6G Location Sensing system. |
| 3 | UVSQ | The OPTI-6G project will enable the UVSQ to explore in depth a new method of localisation and orientation based on RRS/AOA for the OWC, which will enable it to offer numerous training through research projects to young researchers (MSc, PhD and post-doctorate) and to publish numerous research articles. At the same time, the spin-offs of this work can be valorized, for example through technology transfer with companies. In parallel, the OPTI-6G project will lead to the development of mature and advanced hardware/software OWC components (transceivers with and without steering capabilities, MAC and PHY layers for cell-free communication), which will be a major advantage for UVSQ to advance its research on OWCs for various use cases (domestic/industrial indoor networks, automotive communication, space communication), and will lay the foundations for future funded research projects. |
| 5 | REL | Currently RunEL provides its Sparq-2025-P platform of 5G end to end networks at sub 6GHz bands for Vertical Markets and Private Networks such as Stadiums, Smart Factories, AR/VR/XR halls, erc.; The OPTI-6G project will enable RunEL to increase the its product range and diversity by adding additional spectral bands such as photonic bands (1550 nanometer) The exploitation plan will be based on its current worldwide sales and marketing channels as well as via exhibitions ( such as Mobile world Congress-MWC Barcelona) and digital media ( Likedin, Websites, etc,) |
| 6 | OLED | Currently Oledcomm provides its SatelLife, LiFiMAX, SOLERIS product ranges using its current 940nm and 850nm OWC, fully networked technology platforms to the AeroSpace, Defense segments. The OPTI-6G project will enable Oledcomm to increase the its product range and diversity by adding additional spectral bands such as 1550nm and to address new verticals such as industry 4.0 requiring the multibeam and cm localisation functions. The exploitation plan will be based on its current worldwide sales and marketing channels as well as via exhibitions ( CES Las Vegas, Paris Air Show, .... ) and digital media ( Likedin, Websites, etc,) |

### *2.2.2 Dissemination and communication*

***Scientific Publications***

The project partnership has identified target scientific groups at different parts of the economic supply chain: AGVs, XR Headsets, Drones, IoT devices and Robots Manufacturers, and 5G NPN Manufacturers.

- Peer-Reviewed Journals: IEEE transactions, journals and magazines (e.g., Communications, Artificial Intelligence, Automation Science and Engineering, Cognitive Communications and Networking, Image Processing, Human Machine Systems, Industrial Informatics, Industrial Applications, Instrumentation and Measurement, Internet of Things, Robotics, Robotics and Automation, Selected Areas in Communications. Wireless Communications, Antennas and Propagation.
- Scientific Conferences: IEEE Globecom, Association for the Advancement of Artificial Intelligence (AAAI), International Journal of Artificial Intelligence (IJCAI), European Conference on Artificial Intelligence (ECAI), Wireless Communications and Networking Conference (WCNC), Association of Computer Machinary (ACM), European Conference on Antennas and Propagation (EuCAP).
- Industry Conferences: European Conference on Networks and Communications (EuCNC), Mobile World Congress (MWC),
- Trade Fairs & Demo Spaces: Mobile World Congress (MWC), EuCNC & 6G Summit.
- Target Values: Publishing in Peer-reviewed Journal Papers: > 15; Scientific Conference Papers > 30.

***Data-sets Repository***

OPTI-6G project will promote cooperation with the research community by publishing datasets via Zenodo on: KPIs on near IR cell free networks.

***Workshops, Training and special sessions***

Dissemination of expertise oriented towards industrial people through specific industrial-oriented workshops and white papers. The OPTI-6G project is committed to organizing a dedicated workshop in the third year of the project themed on near IR cell free networks.

***Standardization and Regulation Contributions***

OPTI-6G project partners will get involved in the the following standardization groups and make their contributions wherever appropriate:

1. 3GPP Radio Access Working Group
2. ORAN Alliance

***6G IA WGs (the successor to 5G PPP)***

The OPTI-6G project partners will participate in the following work groups; more details of which can be found in: (Work Groups ‹ 5G-PPP). This assumes that each of the WGs will survive for 6G SNS:

1. Technical Board - Presents the overall OPTI-6G system architecture and the proposed Industry 5.0 use cases.
2. Industry 5.0 - Emphasise shifting the focus from the shareholder value to values that reinforce the role and the contribution of industry to society.
3. 5G/Beyond 5G Architecture - Present OPTI-6G architectural concepts and components and foster discussions on KPIs that should be obtained from the architecture depending on the choice of technological trade-offs.
4. Test Measurement and KPI Validation - Present the results of OPTI-6G project laboratory experiments as verification of the performance KPIs that Industry 5.0 is predicting to deliver.
5. Small and Medium Sized Enterprises - Participate in improving the visibility of SMEs in the SNS ecosystem and in general in EU R&D programmes, and ease/promote participation in Networld Europe.

*Alliances, Associations and Academic/Research Networks*
3GPPP Radio Access Working Group
ORAN Alliance
Light Communication Alliance
IEEE 802.15.13 for Industrial Applications
IEEE 802.11bb for Mass-Markets
ITU-T G.hn and G.vlc: From Home Networking to LiFi
Home grid Forum

### *2.2.3 IPR management*

As OPTI-6G will produce significant innovation results because of its originality, business and technological innovations, a strategy handling all the IP Rights matters will be developed under T6.2: Exploitation and Sustainability Plans, creating an encouraging environment for exploitation of the generated creations, discoveries and knowledge.
Concerning exploitation of the intellectual property arising from the project, a detailed IPR management strategy and rules will be proposed and agreed in the consortium agreement prior to the signature of the contract. A preliminary agreement from the partners on the basic principles for managing IP has already been achieved:

- Ownership of background knowledge is not affected by participation and since relevance to the project is included in its definition, it will be made available to the Consortium members free of charge if it is necessary to perform the research in this project.
- Side-ground knowledge (information acquired in parallel to the contract) is to be negotiated between partners on a case-by-case basis if access is needed and relevant for the project.
- Foreground knowledge is owned by the partner generating such information or result.
- Each partner shall make available its foreground knowledge on a royalty-free basis, unless otherwise agreed, to other contractors to the extent that such information is necessary for the production of their own foreground knowledge within the OPTI-6G project.
- Where joint work by several partners has led to generated knowledge and where their respective share of the work cannot be ascertained, they shall have joint ownership of such knowledge. The partners concerned shall agree amongst themselves on the allocation and terms of exercising ownership of that knowledge (otherwise the use of a default regime for joint ownership is foreseen).
- Pre-existing know-how and foreground knowledge will be made available to the other project partners for exploitation purposes at favorable conditions, with respect to the normal commercial conditions applied by the granting partner.
- Research partners will be granted fair compensation in the form of royalties by the partners exploiting the foreground knowledge to which the research partners have contributed.
- Research partners will be entitled to freely reuse internally their foreground knowledge, and will be entitled to freely disseminate such foreground knowledge in any academic paper. The main applicable restriction is that such papers must include as co-authors all the partners who have contributed to such work.
- Research partners will be entitled to create spin-offs for the commercialization of their foreground knowledge, in which case the same conditions would apply to any other partner of the OPTI-6G project.

The Consortium will undertake to preserve the confidentiality of any document, information, knowledge, background know-how or other materials communicated to the partners in relation to the implementation of the project. Where communication of any data, knowledge, foreground or other document is foreseen, the partners in collaboration with the Commission shall first make sure that the recipient will comply with confidentiality obligation and use it only for the purpose for which it is communicated. The confidentiality of any document, information or other material, the disclosure of which could harm, interfere with or otherwise limit the effective protection of their intellectual

property rights, will be maintained during the life of the project. All information made available to other partners will be treated as confidential, unless all parties involved expressly agree to be otherwise. A more precise agreement on IPR issues will be included in the Consortium Agreement. In support of the exploitation and dissemination policies, no additional restraints beyond the standard terms of the Grant Agreement are foreseen. Access rights shall be granted to any of the partners whose personnel have appropriate security clearance upon written request. Except where the partner granting access rights so agrees, such rights shall confer no entitlement to grant sub-licenses. Recipients shall be informed as soon as possible by the partner required to grant access rights of any limitations to the granting of access rights or of any restriction which might substantially affect the granting of access rights.
Table 2.1 lists the access rights (license on patent, other user's rights) to the knowledge created during the project, as a function of the different types of partners. Background knowledge will be required to perform the tasks specified within the project, which will be shared within the Consortium according to the rules stated on Table 2.1.

<table>
<tr><td colspan="2">Table 2.1: Rules for knowledge sharing.</td><td>Access rights to background</td><td>Access rights to foreground</td></tr>
<tr><td rowspan="3">Research and technological development Project OPTI-6G</td><td rowspan="2">For implementing the project</td><td colspan="2">Yes, if a participant needs them for carrying out its own work under the project</td></tr>
<tr><td>Royalty-free, unless otherwise agreed before acceding to the grant agreement</td><td>Royalty-free</td></tr>
<tr><td rowspan="2">For use purposes (exploitation + further research)</td><td colspan="2">Yes, if a participant needs them for using its own foreground knowledge</td></tr>
<tr><td colspan="2">Either royalty-free or on fair and reasonable conditions to be agreed especially when there is a commercial exploitation</td></tr>
</table>

#§COM-DIS-VIS-CDV§#

## *2.3 Summary of specific novel technological solutions for exploitation*

| No | Specific Needs | Expected results | Dissemination, Exploitation and Communication Measures |
|---|---|---|---|
| 1 | Integration of Optical and Wireless Technologies of advanced light related technologies for the design of novel communication schemes, system architectures and protocols, in order to fully integrate these technologies in the communication infrastructure. | 1. Enables additional AOUsto be added without the any need for network planning;<br>2. Reduces the likelihood of NLoS access to RANs/OANs occurring as a result of occluding objects;<br>3. Overcomes bit errors that occur due to Doppler in fast moving OWC channels. | The key towards raising interest is to demonstrate and disseminate in conferences and workshops to: (1) demonstrate the operation of a near IR cell free network; (3) very accurately estimate the mm position and degree orientation of user equipment using near IR sensor array; |
| 2 | Hybrid RF/OWC communications and sensing techniques to underpin selected verticals and applications are in scope. | 1. Accurately measures the location of user devices to 2 - 3mm using Time Difference of Arrival (TDoA), and orientation to 2– 3 degree accuracy using phase of arrival at an near IR sensor array;<br>2. Uses less power measuring distance and orientation thus prolonging battery life capabilities; | |
| 3 | Ultra-low energy and ultra-high capacity solutions for access or end to end connectivity based on optical technologies and their integration within a wireless-optical connectivity continuum. | The OPTI-6G communication system transmits near IR frequencies to multiple UEs thereby producing an ultra-high capacity cell free network solution, whilst the localisation and sensing system is used to locate the UEs in their environment to mm location ad degree orientation accuracy. | |

| | | | |
|---|---|---|---|
| 4 | Design, develop and exhibit the operation of an Radio Optical 6G, so that it can easily be installed in a wide range of different types of properties and where third party application developers can develop and exhibit their innovative network services in homes, businesses and public space buildings using open source development environment. | Adopting a cell free Mobile Network solutions in near IR frequencies will alleviate the competition for spectrum and accelerate the continued growth of hot spots whose current compound annual growth rate (CAGR) is 30% with 628 Million Hotspots recorded in 2022 | |
| 5 | Definition of 5G network architecture and of core technological components | The OPTI-6G project will focus on developing near IR location sensing system for buildings that shows how a location sensing API can be used to develop example Applications and Business Services built from these Localisation Services in Buildings.<br>The OPTI-6G project will contribute to the definition of 6G network architecture and of core technological components that are required within buildings. | |

| No | Target Groups | Outcomes | Impacts |
|---|---|---|---|
| 1 | 5G Infrastructure Association (5G-IA) various 5GPPP WGs or its successor | REL, UBRU will attend the WG meeting and present OPTI-6G concept and results in workshops, white papers and targeted conferences such as EU-CNC | |

| | | | |
|---|---|---|---|
| | Smart Networks and Services Joint Undertaking collaboration (SNS JU) | | |
| 2 | RADAR system: (3GPP TSG SA WG6- Application Enablement and Critical Communication Application) Less than 1cm localisation accuracy field and applications such as: industrial robots, lifting forks, augmented reality glasses and sensors (3GPP TSG SA WG1- Application and System Aspects) Propagation path loss model field (3GPP TSG RAN WG1- Radio Layer 1 and 3GPP TSG RAN WG2 Radio Layer 2 and 3) | REL, will track the 3GPP WG that is concerned with localisation with a view to making an appropriate contribution<br>REL will track the 3GPP WG that is concerned with near IR propagation path loss model with a view to making an appropriate contribution | Potential patents are pending prior to making contribution |
| 3 | Organisations that will witness the lab experiments, use the results of, or provide inputs to the project:<br>• Research Institutions and Academics<br>• Federated Research Infrastructures<br>• Relevant European Technological Platform | Inspire other research groups to reproduce lab experiments in cell Free near IR frequencies to validate or improve on results Potential enhancement of 5GENESIS platform | Promote the application of cell Free Near IR frequencies in communication and sensing |
| 4 | 6G IA<br>3GPPP Radio Access Working Group<br>ORAN Alliance<br>Light Communication Alliance<br>IEEE 802.15.13 for Industrial Applications<br>IEEE 802.11bb for Mass-Markets<br>ITU-T G.hn and G.vlc: From Home Networking to LiFi<br>Home grid Forum | Enhance the existing standards with Photonic near IR Cell Free 5G Network solution | Promotes important technology solution for realising licence free indoor access using Photonic near IR Cell Free 5G Network solution |

# 3 Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#

## *3.1 Work plan and resources*

### 3.1.1 Overall structure of the work plan

The plan is to first collaboratively design the system, then construct it at REL and OLED and finally to ship these solutions to UBRU and UVSQ to perform the system performance, localisation and orientation test experiments and analyse the results. Thus two systems will be constructed. The project is subdivided into three phases that will be executed over three years, namely:

1. Definition and Design Phase
2. Construction Phase
3. Integration and KPI Experimental Testing and Validation Phase

**Year 1 - Definition and Design Phase**

1. Definition of Use Cases and Derivation of User, functional and Technical Requirements
2. Design of near IR Communication and Location Sensing Systems
3. Definition of Testing and KPI Verification Methodology

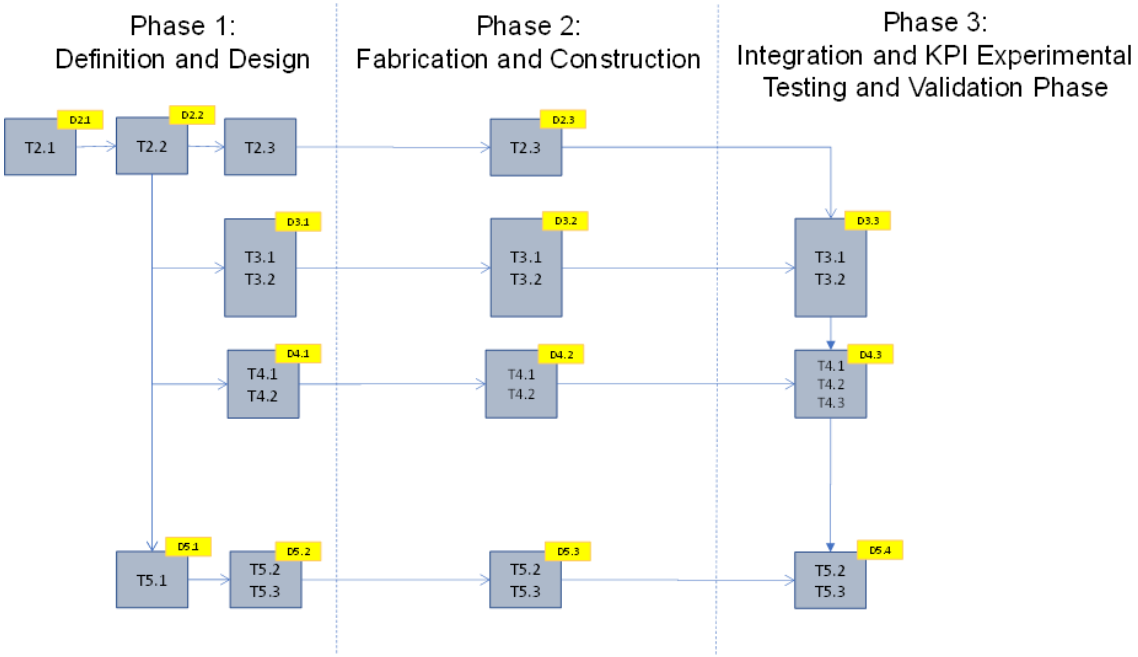
**Year 2 -Construction Phase**

1. Construction of near IR Communication and Location Sensing System

**Year 3 - Integration and KPI Experimental Testing and Validation Phase**

1. KPI Experimental Testing of near IR Communication and Location Sensing System

| Project Phases | | Design and Definition | | | | Construction | | | | Integration and KPI Experimental Testing and KPI Validation | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Work Package | | Year 1 | | | | Year 2 | | | | Year 3 | | | |
| | | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 | Q3 | Q4 |
| 1 | Project and Technical Management | D1.1 | | | D1.2 | | | | D1.3 | | | | D1.4 |
| 2 | User Scenarios, Functional and Technical Requirements, System Architecture | | D2.1 | | D2.2 | | | | D2.3 | | | | |
| 3 | OWC Access Network and User Equipment Design, Integration and Testing | | | | D3.1 | | | | D3.2 | | | D3.3 | |
| 4 | OWC Localisation Sensing | | | | D4.1 | | | | D4.2 | | | D4.3 | |
| 5 | Impact Creation from Exploitation, Dissemination, and Standardization | | D5.1 | | D5.2 | | | | D5.3 | | | | D5.4 |


Phase 1:
Definition and Design
Phase 2:
Fabrication and Construction
Phase 3:
Integration and KPI Experimental
Testing and Validation Phase
T2.1
D2.1
T2.2
D2.2
T2.3
D2.3
T3.1
T3.2
D3.1
D3.2
D3.3
T4.1
T4.2
D4.1
D4.2
T4.3
D4.3
T5.1
D5.1
T5.2
T5.3
D5.2
D5.3
D5.4

### 3.1.2 Graphical Pert Chart Presentation of the Components

WP1: Project and Technical Management - ECOM
T1.1: Project Management- ECOM
T1.2: Technical Management- UBRU
T1.3: Ethics and Data Management- ECOM

WP2: User Scenarios, Functional and Technical Requirements, System Architecture - REL
T2.1: Development of Use Cases and Derivation of User, functional & Technical Requirements- REL
T2.2: Design of Optical Communication and Sensing Architecture - UVSQ
T2.3: Testing and KPI Verification Methodology –OLED

WP3: OWC Access Network and User Equipment Design, Integration and Testing - OLED
T3.1: near IR OWC Access Network and User Equipment Design & Development- OLED
T3.2: Integration and Testing of Radio-Optical Communication System - REL

WP4: OWC Localisation Sensing - UBRU
T4.1: OWC TDoA Localisation Sensing- REL
T4.2: OWC RSS and AoA Localisation Sensing- UVSQ
T4.3: Integration and Testing of Optical Location Systems - UBRU

WP5: Impact Creation from Exploitation, Dissemination and Standardisation - REL
T5.1: Technoeconomic Analysis and Exploitation- OLED
T5.2: Dissemination Activities - REL
T5.3: Co-ordination of liaisons with regulatory, industry associations and standardization bodies - OLED

### 3.1.3 Work packages and their components

| | | Year | | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Month | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| | | | | | | | | | | | | | | | M1 | | | | | | | | | | | | M2 | | | | | | | | | | | M3 |
| WP1 | Project and Technical Management | EUR | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T1.1 | Administrative and Project Management | EUR | | | | D1.1 | | | | | | | | | D1.2 | | | | | | | | | | | | D1.3 | | | | | | | | | | D1.4 |
| T1.2 | Technical Management | UBRU | | | | | | | | | | | | | D1.2 | | | | | | | | | | | | D1.3 | | | | | | | | | | D1.4 |
| T1.3 | Data management and ethics | EUR | | | | D1.1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| WP2 | User Scenarios, Functional and Technical F | REL | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T2.1 | Development of Use Cases and Derivation of User, functional and Technical Requirements | REL | | | | | | | D2.1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T2.2 | Design of Optical Communication and Sensing Architecture | UVSQ | | | | | | | | | | | | D2.2 | | | | | | | | | | | | | | | | | | | | | | | |
| T2.3 | KPI Testing and Verification Methodology | OLED | | | | | | | | | | | | | | | | | | D2.3 | | | | | | | | | | | | | | | | | |
| WP3 | OWC Access Network and User | OLED | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T3.1 | OWC Access Network and User Equipment Design Design & Development | OLED | | | | | | | | | | | | | D3.1 | | | | | | | | | | | | D3.2 | | | | | | | | | | D3.3 |
| T3.2 | Integration and KPI Testing of Optical Communication System | REL | | | | | | | | | | | | | D3.1 | | | | | | | | | | | | D3.2 | | | | | | | | | | D3.3 |
| WP4 | OWC Localisation Sensing | UBRU | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T4.1 | mmW & OWC TDoA Localisation Sensing | REL | | | | | | | | | | | | | D4.1 | | | | | | | | | | | | D4.2 | | | | | | | | | | D4.3 |
| T4.2 | OWC RSS and AoA Localisation Sensing | UVSQ | | | | | | | | | | | | | D4.1 | | | | | | | | | | | | D4.2 | | | | | | | | | | D4.3 |
| T4.3 | Integration and Testing of Optical Location Systems | UBRU | | | | | | | | | | | | | D4.1 | | | | | | | | | | | | D4.2 | | | | | | | | | | D4.3 |
| WP5 | Impact Creation from Exploitation, Dissemination and Standardisation | REL | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T5.1 | Technoeconomic Analysis and Exploitation | OLED | | | | | | | D5.1 | | | | | | D5.2 | | | | | | | | | | | | D5.3 | | | | | | | | | | D5.4 |
| T5.2 | Dissemination Activities | REL | | | | | | | | | | | | | D5.2 | | | | | | | | | | | | D5.3 | | | | | | | | | | D5.4 |
| T5.3 | Liaisons with industry asspciations, regulatory and standardization bodies | OLED | | | | | | | | | | | | | D5.2 | | | | | | | | | | | | D5.3 | | | | | | | | | | D5.4 |

### 3.1.3 Detailed work package description

| WP No | 1 | Lead beneficiary | | ECOM | |
|---|---|---|---|---|---|
| WP Title | Project Management | | | | |
| Partner No | | 2 | 3 | 4 | 5 |
| Short name | ECOM | UBRU | UVSQ | REL | OLED |
| PMs: | 21 | 14 | 0 | 0 | 0 |
| Start mth | M01 | End month | M36 | | |

Objectives: This work package is responsible for project management, which includes project coordination, administrative project management, scientific and technical management, coordination of all WPs, and the management of the quality assurance process for the project output. These activities are performed through the implementation of the project management structure and procedures, some of which are described in section 2.2. WP objectives are:

- To establish efficient operation of all project bodies and work of the project participants, including proper decision making and conflict resolution at all levels;
- To ensure required quality of project work and its results as well as to perform self-assessment and corresponding project monitoring, including risk management;
- To organize the overall project administration and supervision of financial flows as well as communication with EC and project reporting;
- To ensure the quality and soundness of the scientific and technical work performed in the project through appropriate scientific and technical management activities;
- To provide the necessary collaborative environment for the consortium.

Efficient project management, provided by this WP will ensure that all scientific, technical and administrative activities, as defined in the project plan, are executed to a high-quality level, and that all deliverables, milestones and reports are completed according to the project schedule in time and quality.

**Task 1.1 Administrative and Project Management:** (Duration M1-M36) (Lead: ECOM) (Participants: )
This task will provide the overall project administration by performing self-assessment activities (resources planning /distribution, project monitoring, corrective actions if needed), deliverables quality assurance, risk management, and external factors consideration. Work of the project management bodies will be enabled within T1.1, including maintenance of the decision-making process and conflict management for technical and other related aspects. An appropriate communication and collaborative environment for the consortium members (e.g., mailing lists, project repository, reporting tools) will also be set up. ECOM will serve as the Project Coordinator: Adam Kapovits who is a well-recognized project manager and an experienced scientist from academia UBRU (John Cosmas) will act as science manager who has a extensive experience in EU Framework Programmes at both technical and administrative level and were involved in several projects, including IoRL as Principal Investigator, WP leader and Project Coordinator. Currently Adam Kapovits The Project Coordinator will be assisted by the Project Coordination Committee (PCC), which will be the formal decision-making body for the project and consists of one delegate (Partner Representative) from each project partner. Normally, PCC meetings will be called in association with Technical Meetings. Participation by full partners or their delegates is mandatory. The chairperson of the PCC is the PC. The PCC shall meet as required, between four to six times per year. The responsibility and decision-making will be outlined in the Consortium Agreement.

**Task 1.2 Technical Management** (Duration M1-M36) (Lead: UBRU) (Participants: )
This task covers the overall coordination of the technical and scientific activities during development, integration, and piloting of the project. The main role of TM is to facilitate smooth cooperation between partners developing different elements of the platform, to ensure that they conform to the established architecture and interfaces and to verify that the developed software is properly tested and documented, to ensure timely completion and quality assurance of the technical deliverables. The TM will coordinate the technical activities and oversee them via regular meetings (physical or virtual) with the partners.

**Task 1.3 Ethics and Data Management** (Duration M1-M06) (Lead: ECOM) (Participants: UVSQ)
This task will investigate, design, and monitor the procedures and protocols necessary for handling legal issues during the project's lifetime. It will also set out the ethics requirements that the project must comply with (e.g., procedures, criteria to identify/recruit research participants, templates for informed consent, etc.) and manage the ethical issues concerning with the design of the information and consent forms that will be used in case of involvement of humans through use case validations, whilst ensuring General Data Protection Regulation (GDPR) and Privacy Compliance Policy. Preparation of the ethical, legal guidelines and the research ethical protocol as part of deliverable D1.1, stipulating the ethical principles that the consortium partners will adhere to in their work. Gender balance will be monitored throughout the project to ensure that the female/male ratio is maintained as close to 1:1 as possible for people employed during its course.

D1.1: Ethics and Data Management Action Plan Report (M06), (R) [ ECOM] – describes how ethics and data management issues will be dealt with in the project
D1.2: 1st Period Management Report(M12), (R) [ECOM] – presents the technical and financial progress of the project after the first year
D1.3: 2nd Period Management Report (M24), (R) [ECOM] - presents the technical and financial progress of the project after the second year
D1.4: 3rd Period Management Report (M36), (R) [ECOM] - presents the technical and financial progress of the project at the end of the project

| WP No | 2 | Lead beneficiary | | REL | |
|---|---|---|---|---|---|
| WP Title | User Scenarios, Functional and Technical Requirements, System Architecture | | | | |
| Partner No | 1 | 2 | 3 | 4 | 5 |
| Short name | ECOM | UBRU | UVSQ | REL | OLED |
| PMs: | 0 | 9 | 3 | 30 | 12 |
| Start mth | M01 | End month | M18 | | |

Objectives: The Objective of WP-2 is to derive the technical requirements and specifications of the overall system and the different components of the hybrid Photonic-Microwave Cell free Network.
OPTIC-6G team will take the Up to Bottom approach by starting from analysing the ultimate requirements of use case scenarios for in-building or indoor Communication and Sensing applications and services.
Following the definition of the proposed solution specifications and requirements, the team will proceed with the detailed design of the Optical Domain Cell Free Networks that will be tested in the laboratory environment.
In parallel the team will develop a comprehensive testing and verification methodology of the different KPI that will be measured and analysed in the lab tests.

**Task 2.1** Development of Use Cases and Derivation of User, functional and Technical Requirements: (Duration M1-M6) (Lead: REL (Participants: UBRU, UVSQ, OLED). This task will analyse the requirements of different indoor use cases such as in-building navigation for people, robots and drones and derive from the main specifications and KPIs for the Photonic Cell Free Network.

Task 2.2 Design of Optical Communication and Sensing Architecture: (Duration M4-M12) (Lead: UVSQ) (Participants: UBRU)
This task covers the design of interfaces with optical emitter and receiver and the design of the optical part of the communications with optimization of parameters (data rate, coverage, sensitivity of the receivers). The integration of the sensing part and 3D localization with AoA measurements will be included also in the architecture of the transceiver.

Task 2.3 Testing and KPI Verification Methodology: (Duration M10-M18) (Lead: OLED) (Participants: UBRU, UVSQ, REL)

This task will provide guidelines for the testing configurations and methodologies to be used for the characterization and validation of the performance KPIs in the scope of WP3 and WP4. We will define the different test beds and experimental set-ups to be used in the labs to measure the KPIs defined in T2.1 considering the definitions of the architecture in T2.2. This task will progress in parallel with T2.1 and T2.2 and finalize in M24 with the deliverable D2.3.

D2.1: **OWC Cell Free Network** Use Cases and User, functional and Technical Requirements (M06), (R) [REL] – Describes and analyses the potential Use Cases that the Photonic Cell Free Network is intended to provide solutions for. Following the Use Cases analysis the document will provide the proposed system overall specifications as well as the specifications for its main modules and technologies.
D2.2: OWC Cell Free Network Architecture (M12), (R) [UVSQ] – presents the description of the validated architectures that have been discussed and the comments about the hypothesis and contexts of using.
D2.3: OWC Cell Free Network Testing and KPI Verification Methodology (M18), (R) [OLED] - presents the description of the multiple test-beds and experimental set-ups to validate the performance of the different KPIs. The document will present a detailed description of the experimental set-ups in the lab and the expected results.

| WP No | 3 | Lead beneficiary | | | OLED |
|---|---|---|---|---|---|
| WP Title | OWC Access Network and User Equipment Design, Integration and Testing | | | | |
| Partner No | 1 | 2 | 3 | 4 | 5 |
| Short name | ECOM | UBRU | UVSQ | REL | OLED |
| PMs: | 0 | 6 | 5 | 6 | 36 |
| Start mth | M01 | | End month | M36 | |

Objectives:
This work package is intended to design, develop and build the OWC equipment and the integration with RUNEL Cell Free Network with AI Scheduler.
Following the definition of the proposed solution specifications and requirements, the team will proceed with the detailed design of the Optical Domain Cell Free Networks that will be tested in the laboratory environment.
In parallel the team will develop a comprehensive testing and verification methodology of the different KPI that will be measured and analysed in the lab tests.

Task 3.1 OWC Access Network and User Equipment Design & Development: (Duration M1-M36) (Lead: OLED) (Participants: UVSQ)
This task consists in the design and the development of 2 sets of OWC systems including for each of them 4 OWC transceivers configured as Access points and 1 of the configured as end point (end user). 4 OWC transceivers with the following characteristics :

- Near IR (1550nm) light source
- VCSEL phase array (allowing beam steering techniques to be incorporated)
- bidirectional Gbps transmission rate depending on opening angle
- 5 to 25 degrees opening angle over a distance of 10m
- Software layers for multi user management
- Handover functions software layers included

1 OWC system for the user side

Task 3.2 Integration and KPI Testing of Optical Communication System: (Duration M1-M36) (Lead: REL) (Participants: UBRU, UVSQ, OLED)
This task consists in integrating in the transceiver location systems from Brunel and UVSQ and measure the following KPIs :

- Speed rate distribution over the volume covered
- Bit error rate distribution over the volume covered
- SNR for the OWC distribution over the volume covered

- olatency for both downlink and uplink
- User location accuracy for both downlink and uplink

D3.1: Near IR Cell Free Communication System Design (M12), (R) [OLED] – Following the Use Cases analysis the document will provide the near IR system overall specifications as well as the specifications for its main building blocks and technologies.

D3.2: Near IR Communication System Construction (M24), (R) [UVSQ] – presents the 2 sets of 4 OWC transceivers + 1 OWC end points that will be sent to Oledcomm and RUNEL labs

D3.3: Near IR Cell Free Communication System Integration, Test and KPI Validation (M36), (R) [REL] - presents the overall

| WP No | 4 | Lead beneficiary | | UBRU | |
|---|---|---|---|---|---|
| WP Title | OWC Localisation Sensing | | | | |
| Partner No | 1 | 2 | 3 | 4 | 5 |
| Short name | ECOM | UBRU | UVSQ | REL | OLED |
| PMs: | 0 | 16 | 50 | 4 | 0 |
| Start mth | M01 | | End month | M36 | |

Objectives: This work package designs and develops algorithms for obtaining location and orientation of User Equipment in Photonic OWC network and performs measurements on constructed experimental test benches to test Key Performance Indicators and analyses the results. If necessary modifications of the algorithms may be required because of their operation in the Photonic OWC as opposed to radio networks.

**Task 4.1** OWC TDoA Localisation Sensing: (Duration M1-M36) (Lead: REL (Participants: UBRU, UVSQ, )
This task covers the development of the algorithms and experimental setup for the estimation of the 3D position and orientation of the UE by measuring the TDoA of the received near IR signals. Two experimental Cell Free test benches with localization infrastructure will be constructed to perform the integration between the Microwave Cellfree Network and the OWC links, one at the RunEL premises in Israel and the second in Oledcom premises in France,

Task 4.2 OWC RSS and AoA Localisation Sensing: (Duration M1-M36) (Lead: UVSQ) (Participants: )
This task covers the development of the algorithms and experimental setup for the estimation of the 3D position and orientation of the UE by measuring the AoA and RSS of the received optical signal. Simulations will be carried out to validate the principles of the proposed solution and to optimise its parameters, and then an experimental test bench will be built to evaluate its performance in real situations.

Task 4.3 Integration and Testing of Optical Location Systems: (Duration M1-M36) (Lead: UBRU) (Participants: UVSQ)
This task performs the experimental procedures and measures the position of the User Equipment as defined in Task 2.3 Testing and KPI Verification Methodology; It then analyses the results. The test will initially be done in RunEL and Oledcom labs and following the System integration and initial performance testing the setups will be transfered to the BrunEL University and to Universite de Versailles to perform the UE localization experiments,

D4.1: OWC Localisation Sensing System Design (M12), (R) [REL] – describes the design of the localisation sensing systems
D4.2: OWC Localisation Sensing System Construction (M24), (R) [UVSQ] – presents the Construction of the localisation systems
D4.3: OWC Localisation Sensing System Integration, Test and KPI Validation (M36), (R) [UBRU] - presents the outcome of the performed the experimental procedures, the measurement results obtained and analyses the results.

| WP No | 5 | Lead beneficiary | | REL | |
|---|---|---|---|---|---|
| WP Title | Impact Creation from Exploitation, Dissemination and Standardisation | | | | |
| Partner No | 1 | 2 | 3 | 5 | 6 |
| Short name | ECOM | UBRU | UVSQ | REL | OLED |
| PMs: | 10 | 3 | 2 | 6 | 10 |
| Start mth | M01 | End month | M36 | | |

Objectives: This WP is responsible for external impact creation. Its objectives are detailed as follows:
• Dissemination – Creation of impact of the project's research results in the European and global research and business community as well as among other stakeholders and the interested public. Dissemination includes the regular internal evaluation by its industry participants
• Standardization – Support the development of contributions to pre-standardization initiatives as well as standardization.
• Techno-economic analysis of OPTI-6G main case applications including user needs.
• Exploitation – Developing plans to support commercial exploitation of the results as well as executing activities for securing Intellectual Property Rights.
• Programme level coordination within the 6G IA, 3GPP, ORAN Alliance, Light Communication Alliance, IEEE 802.15.13 for Industrial Applications, IEEE 802.11bb for Mass-Markets, ITU-T G.hn and G.vlc: From Home Networking to LiFi, Home grid Forum

Task 5.1 Technoeconomic Analysis and Exploitation: (Duration M1-M36) (Lead: OLED) (Participants: REL)
This task will develop the strategy for exploiting the results. In month 6 the clear potential impact of the system based on the technical description will be defined. At the conclusion of the project, technological economic analysis will be published defining the technical features that we will have implemented. Individual partner plans will be elaborated on how the development of Photonic Cell Free 5G Network that can be exploited by their organisations. The economic impact of the project results on this B5G and 6G will be analysed.

Task 5.2 Dissemination Activities: (Duration M1-M36) (Lead: REL) (Participants: ECOM, UBRU, UVSQ, OLED)
This task organises the transfer of knowledge and project results to the outside world, and within the consortium as outlined in section 2.2.2 Dissemination and Communication. It will report on progress in deliverables 5.2, 5.3, 5.4

Task 5.3 Co-ordination of liaisons with regulatory, industry associations and standardization bodies: (Duration M1-M06) (Lead: OLED) (Participants: ECOM)
This task involves coordinating and maintaining communication with relevant regulatory bodies, industry associations, and standardization bodies in the communications sector. The purpose of this task is to ensure that the project team is up-to-date with any regulatory or industry requirements, and that the project's cell free photonic 5G network technology aligns with any relevant standards. The task will involve identifying and contacting key stakeholders in the industry, such as regulatory bodies responsible for health and safety, and industry associations representing the indoor networking sector.

| |
|---|
| The project team will work to establish productive relationships with these stakeholders, and to keep them informed about the project's progress and any potential impacts on their work. In addition, the task will involve monitoring and analyzing any changes in regulatory or industry requirements that could impact the project, and communicating these changes to the project team. The team will then assess any necessary modifications to the project plan or technology to ensure compliance with these requirements. Finally, the task will also involve engaging with relevant standardization bodies to ensure that the project's cell free photonic network, localisation and orientation measurement technologies align with any applicable standards. This could involve reviewing and modifying project deliverables, such as training materials or inspection reports, to meet these standards. |
| D5.1: **Technoeconomic Analysis and Exploitation Plans** (M06), (R) [ OLED] – The aim is to analyse the Techno-economic aspects of the research performed in the OPTI-6G project to guide the beneficiaries to answer the following questions: (1) What are the areas that OPTI-6G can provide a substantial impact to the markets? (2) What are the products and services needed? The analysis will be focused on the main use-cases identified by the OPTI-6G consortium.<br>D5.2:First year Dissemination, and liaisons with regulatory, industry associations and standardization bodies activities (M12), (R) [REL] – Provides report of project dissemination activities year 1.<br>D5.3:Second year Dissemination, and liaisons with regulatory, industry associations and standardization bodies activities (M24), (R) [REL] - Provides report of project dissemination activities year 2.<br>D5.4:Final year Dissemination, Exploitation activities and liaisons with regulatory, industry associations and standardization bodies (M36), (R) [REL] - Provides report of project dissemination activities year 3. |

### *3.1.4 Summary of staff effort*

| Participant Number/Short Name | WP1 | WP2 | WP3 | WP4 | WP5 | Total Person-Months per Participant |
|---|---|---|---|---|---|---|
| 01 ECOM | 21 | 0 | 0 | 0 | 10 | 31 |
| 02 UBRU | 14 | 9 | 2 | 16 | 3 | 44 |
| 03 UVSQ | 0 | 3 | 5 | 50 | 2 | 60 |
| 04 REL | 0 | 30 | 6 | 4 | 6 | 46 |
| 05 OLED | 0 | 12 | 36 | 0 | 10 | 58 |

### *3.1.5 List of deliverables*

| #[1] | Deliverable name | WP No | Lead | Type[2] | Diss Level[3] | Delivery Date[4] |
|---|---|---|---|---|---|---|
| D1.1 | Ethics and Data Management Action Plan Report | 1 | ECOM | DMP | PP | M06 |
| D1.2 | 1st Period Management Report | 1 | ECOM | R | PP | M12 |
| D1.3 | 2nd Period Management Report | 1 | ECOM | R | PP | M36 |
| D1.4 | 3rd Period Management Report | 1 | ECOM | R | PP | M36 |
| D2.1 | OWC Cell Free Network Use Cases and User, functional and Technical Requirements | 2 | REL | R | PU | M06 |
| D2.2 | OWC a Cell Free Network Architecture | 2 | UVSQ | R | PU | M12 |
| D2.3 | OWC Cell Free Network Testing and KPI Verification Methodology | 2 | OLED | R | PU | M18 |
| D3.1 | Near IR Cell Free Communication System Design | 3 | REL | R/P | PU | M12 |
| D3.2 | Near IR Cell Free Communication System Construction | 3 | OLED | R/P | PU | M24 |
| D3.3 | Near IR Cell Free Communication System Integration, Test and KPI Validation | 3 | REL | R/P | PU | M36 |
| D4.1 | OWC Localisation Sensing System Design | 4 | REL | R/P | PU | M12 |
| D4.2 | OWC Localisation Sensing System Construction | 4 | UVSQ | R/P | PU | M24 |
| D4.3 | OWC Localisation Sensing System Integration, Test and KPI Validation | 4 | UBRU | R/P | PU | M36 |
| D5.1 | Technoeconomic Analysis and Exploitation Plans | 5 | OLED | R | PU | M06 |
| D5.2 | First year Dissemination, liaisons with regulatory, industry associations and standardization bodies activities | 5 | REL | R | PU | M12 |
| D5.3 | Second year Dissemination, liaisons with regulatory, industry associations and standardization bodies activities | 5 | REL | R | PU | M24 |
| D5.4 | Final year Dissemination, liaisons with regulatory, industry associations and standardization bodies activities | 5 | REL | R | PU | M36 |

### *3.1.6 List of Milestones*

| MS No | Milestone name | Related WP(s) | Estimated date | Means of verification[1] |
|---|---|---|---|---|
| 1 | Completion of Definition and Design Phase | WP2, 3, 4 | M12 | D2.1, D2.2, D3.1, D4.1, D5.1 |
| 2 | Completion of Construction Phase | WP3, 4 | M24 | D2.3, D3.2, D4.2 |
| 3 | Completion of Integration and KPI Experimental Testing and Validation Phase | WP 2, 3, 4 | M36 | D3.3, D4.3 |

### *3.1.7 Critical risks for implementation* #@RSK-MGT-RM@#

| WP | Risk | Probability | Risk Level | Mitigation and Contingency Plan |
|---|---|---|---|---|
| 1 | What happens if UK does not agree with EU on funding projects | L | H | Find some other organisation |
| 1 | Prohibitive cost of licensing multiple frequency bands | L | M | Mitigating measure is to select bandwidths in both licenses and unlicensed frequency bands |
| 2 | Another technology provider develops a similar solution to what is being proposed in the OPTI-6G project | L | H | Invite the technology provider to a OPTI-6G project workshop to identify differences and similarities between two solutions and to organise collaboration on joint standardisation efforts. |
| 2 | 6G performance is not sufficiently good enough for the demanding industry 4.0 use cases | M | L | Investigate method to improve the performance and recommend future research areas to resolve the problem |
| 3 | Interfacing and radio and optical system faces some unknowns that need to be resolved wrt to the carrier IF frequency. | H | L | If there are problems with this a baseband signal may be required from the RUNEL RU which can provide any frequency between 0 and 5 GHz |
| 3 | Using the 1550 nm Vcsels for multi user fully networked optical wireless connectivity | H | L | An in-depth study will be preformed in the start of the project to discover how to use 1550 nm to fine tune the circuit design to operate with this wavelength |
| 4 | The PN Sequence used for measuring delay in the near IR TDoA measurement system is to noisy | L | L | Investigate methods to reduce interference, and or reduce the size of the testing area, and or increase the sequence lngth |
| 4 | Performance predicted in simulations cannot be reproduced in equivalent laboratory experiments | L | L | Investigate method to improve the laboratory test setup performance and recommend future research areas to resolve the problem |
| 5 | project developments don't improve, or drive towards the overall call's targeted outcomes | L | L | Perform alignment during project scoping and throughout execution. |
| 5 | disseminating and exploitation not communicated with industry to drive impact | L | L | dissemination will be driven over the 3 years of the project across multiple communication methods. |

#§RSK-MGT-RM§#

## *3.2 Construction of consortium as a whole* #@CON-SOR-CS@# #@PRJ-MGT-PM@#

### *3.2.1 Beneficiaries Contributions*

| Technology | EUR | UBRU | UVSQ | REL | OLED |
|---|---|---|---|---|---|
| Project Management | X | | | | |
| Technical Management | | X | | | |
| Cell Free Network with AI Scheduler | | | | X | |
| High Speed Photodetectors and Transducer | | | | | X |
| Optical Vcsel Phased Array Emitters/Sensors | | | | | X |
| Localisation Sensing from near IR Sensors | | X | X | X | X |
| | | | | | |

### *3.2.2 Countries Involved*

| Country | Beneficiary 1 | Beneficiary 2 | Total MM |
|---|---|---|---|
| Germany | EUR | | 31 |
| Israel | REL | | 46 |
| France | OLED | UVSQ | 118 |
| United Kingdom | UBRU | | 44 |

### *3.2.3 Beneficiaries Types*

We have in this project 5 beneficiaries of the following types:

- 2 SMEs - REL and OLED
- 2 Academic partners: UBRU and USVQ
- 1 Industrial partner EUR

### *3.2.4 Allocation of resources*

A full system will be integrated at RUNEL and OLED and then shipped to UBRU and UVSQ for performing localisation experiments.

In the following we provide a detailed breakdown of the other costs, where they exceed 15% of the budgeted personnel costs.

| **Participant P02 UBRU** | **Cost (€)** | **Justification** |
|---|---|---|
| Amerisoft Core | 40,000 | Required to complete building of 5G Network |
| High Performance Server | 8,000 | Required for hosting Amerisoft Core |
| 2 x 5G User Equipment | 2,000 | Required for receivers |

| | | |
|---|---|---|
| **Other goods and services** | 25,000 | Required for 4 Plenary meeting per year, 2 to 3 project review meeting with EU and integration meeting for equipment. |
| | 5000 | Consumables related to setting up the experimental laboratory for making measurements |
| **Total** | 80,000 | |

| **Participant04: REL** | **Cost (€)** | **Justification** |
|---|---|---|
| Travel Expenses | 30000 | Participation in Quarterly General Assembly meetings and Installation and integration of the cel lfree test bed demo |
| | | |
| **Other goods and services** | 172000 | Cost of 2 x Cell free networks systems for research development and proof of concept lab tests including Core + CU + DU + 4 x RUs with localization accelerators + Localization Servers for each cellfree network |
| **Total** | 202000 | |

| **Participant03: UVSQ** | **Cost (€)** | **Justification** |
|---|---|---|
| Amerisoft Core | 40,000 | Required to complete building of 5G Network |
| High Performance Server | 8,000 | Required for hosting Amerisoft Core |
| 2 x 5G User Equipment | 2,000 | Required for receivers |
| **Travel** | 25,000 | Required for 4 Plenary meeting per year, 2 to 3 project review meeting with EU and integration meeting for equipment. |
| **Other goods and services** | 20,000 | Consumables related to setting up the experimental laboratory for making measurements |
| **Total** | 95,000 | |

### *3.2.5 In-Kind contributions to Operational Activities (IKOP)*

| **Participant Number** | **Short Name** | **Corresponding 6GIA Member (Legal Entity Name)** | **Cost (€)** |
|---|---|---|---|
| **1** | EUR | EURESCOM-EUROPEAN INSTITUTE FOR RESEARCH AND STRATEGIC STUDIES IN TELECOMMUNICATIONS GMBH | 45,187.50 |
| **2** | UBRU | BRUNEL UNIVERSITY LONDON | no IKOP contribution since not for profit |
| **3** | UVSQ | UNIVERSITE DE VERSAILLES SAINT-QUENTIN-EN-YVELINES | no IKOP contribution since not for profit |

| | | | |
|---|---|---|---|
| **4** | REL | RUNEL NGMT LTD | 69,700.00 |
| **5** | OLED | OLEDCOMM SAS | 60,300.00 |
| **Total IKOP costs: (€)** | | | **175,187.50** |
| **Total IKOP contribution in percentage of the overall eligible costs: 6.43%** | | | |

#§QUA-LIT-QL§# #§WRK-PLA-WP§#

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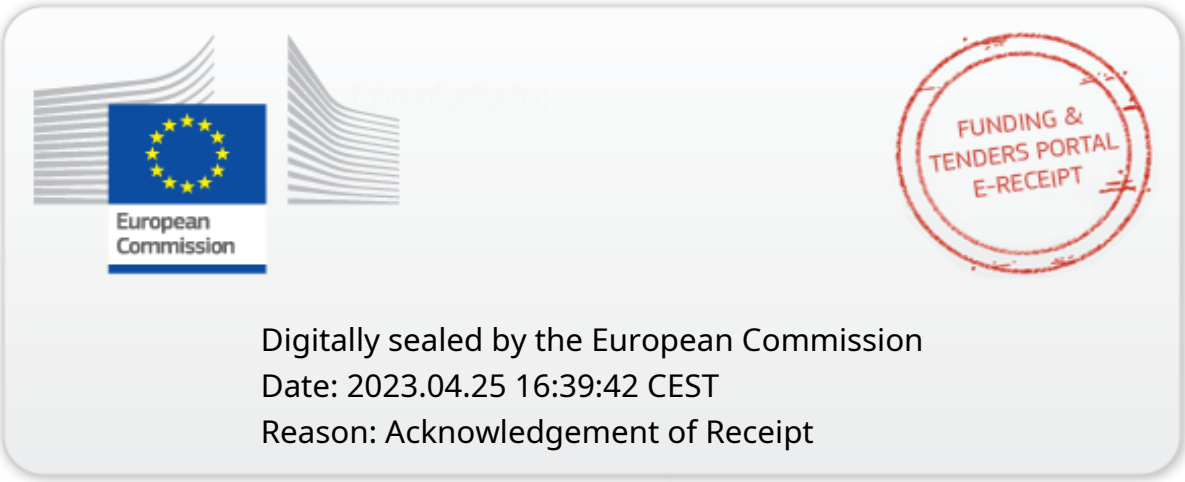
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